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| Title:<br>SYSTEM IMPACT STUDY REPORT                                              |              | Issue Date: | 10-04-2019 |

## MISO DPP April 2018 East (ATC) Phase 1

### System Impact Study Report

**J986, J1000, J1002, J1003, J1009, J1010, J1011, J1042, J1044,  
J1047, J1051, J1053, J1085, J1101, J1104, J1121, J1127, J1129, J1153,  
J1154, J1171, J1183, and J1188**

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## **1.0 EXECUTIVE SUMMARY**

Twenty-three (23) resource projects have requested to interconnect to the MISO transmission network in the East (ATC) Area and are included in the Definitive Planning Phase 2018 April Phase 1 study (Apr 18 DPP Phase 1). All Generating Facilities, except one, have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS). J1183 has only requested ERIS.

This report presents the steady-state study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generator interconnection requests in the Apr 18 DPP Phase 1 study. No stability, short circuit or PJM affected system analysis were performed in Phase 1. The study was performed under the direction of Midcontinent Independent System Operator (MISO) by ATC and an ad hoc study group. The results for 2023 scenario are summarized below.

### **1.1 Project List**

The Apr 18 DPP Phase 1 has twenty-three (23) interconnection requests with a combined nameplate rating of 3,610.01 MW. The Apr 18 DPP Phase 1 generator interconnection requests are listed in Table 1.1-1. The one-line diagrams of the interconnection facilities are shown in Appendix C.

Table 1.1-1 – Generating Facilities in DPP April 2018 Phase 1 East (ATC) Area

| MISO Queue # | Service Type | Control Area | County, State              | Point of Interconnection        | Fuel Type | Requested MW | Dispatch (MW) <sup>1</sup> |        |                    |        |
|--------------|--------------|--------------|----------------------------|---------------------------------|-----------|--------------|----------------------------|--------|--------------------|--------|
|              |              |              |                            |                                 |           |              | Batteries Discharging      |        | Batteries Charging |        |
|              |              |              |                            |                                 |           |              | 23SH                       | 23SUM  | 23SH               | 23SUM  |
| J986         | NRIS         | ALTE         | Wood, WI                   | Port Edwards - Sand Lake 138-kV | Solar     | 149.76       | 74.88                      | 149.76 | 74.88              | 149.76 |
| J1000        | NRIS         | ALTE         | Grant, WI                  | Nelson Dewey 138-kV             | Solar     | 50           | 25                         | 50     | 25                 | 50     |
| J1002        | NRIS         | ALTE         | Waushara, WI               | Wautoma 138-kV                  | Solar     | 99           | 49.5                       | 99     | 49.5               | 99     |
| J1003        | NRIS         | ALTE         | Dodge, WI                  | North Beaver Dam 69-kV          | Solar     | 50           | 25                         | 50     | 25                 | 50     |
| J1009        | NRIS         | WEC          | Kenosha, WI                | Pleasant Prairie-Racine 345-kV  | Solar     | 400          | 200                        | 400    | 200                | 400    |
| J1010        | NRIS         | WEC          | Kenosha, WI                | Arcadian-Zion 345-kV            | Solar     | 400          | 200                        | 400    | 200                | 400    |
| J1011        | NRIS         | WEC          | Kenosha, WI                | Arcadian-Zion 345-kV            | Solar     | 400          | 200                        | 400    | 200                | 400    |
| J1042        | NRIS         | ALTE         | Walworth, WI               | North Lake Geneva 138-kV        | Solar     | 200          | 100                        | 200    | 100                | 200    |
| J1044        | NRIS         | ALTE         | Walworth, WI               | North Lake Geneva Tap 138-kV    | Solar     | 200          | 100                        | 200    | 100                | 200    |
| J1047        | NRIS         | WEC          | Walworth, WI               | North Lake Geneva Tap 138-kV    | Solar     | 200          | 100                        | 200    | 100                | 200    |
| J1051        | NRIS         | WEC          | Jefferson, WI              | Concord 138-kV                  | Battery   | 50           | 50                         | 50     | -50                | -50    |
| J1053        | NRIS         | WEC          | Jefferson, WI              | Concord 138-kV                  | Solar     | 300          | 150                        | 300    | 150                | 300    |
| J1085        | NRIS         | WPS          | Marathon, Taylor, Wood, WI | Gardner Park-Stone Lake 345-kV  | Wind      | 300          | 300                        | 47.1   | 300                | 47.1   |
| J1101        | NRIS         | WPS          | Manitowoc, WI              | Kewaunee 138-kV                 | Battery   | 20           | 20                         | 20     | -20                | -20    |
| J1104        | NRIS         | WEC          | Washington, WI             | Germantown-Bark River 138-kV    | Battery   | 100          | 100                        | 100    | -100               | -100   |
| J1121        | NRIS         | WPS          | Marathon, WI               | Gardner Park-Stone Lake 345-kV  | Solar     | 200          | 100                        | 200    | 100                | 200    |
| J1127        | NRIS         | ALTE         | Green, WI                  | Albany-Bass Creek 138-kV        | Solar     | 49.9         | 24.95                      | 49.9   | 24.95              | 49.9   |
| J1129        | NRIS         | ALTE         | Rock, WI                   | Paddock-Townline Rd 138-kV      | Solar     | 65           | 32.5                       | 65     | 32.5               | 65     |
| J1153        | NRIS         | WEC          | Sheboygan, WI              | Holland 138-kV                  | Solar     | 150          | 75                         | 150    | 75                 | 150    |

|       |      |      |               |                       |       |      |       |      |       |      |
|-------|------|------|---------------|-----------------------|-------|------|-------|------|-------|------|
| J1154 | NRIS | WEC  | Jefferson, WI | Jefferson 138-kV      | Solar | 75   | 37.5  | 75   | 37.5  | 75   |
| J1171 | NRIS | WEC  | Dodge, WI     | Butternut 138-kV      | Solar | 100  | 50    | 100  | 50    | 100  |
| J1183 | ERIS | MIUP | Delta, MI     | Garden Corners 138-kV | Solar | 1.35 | 0.675 | 1.35 | 0.675 | 1.35 |
| J1188 | NRIS | ALTE | Rock, WI      | Sheepskin 69-kV       | Solar | 50   | 25    | 50   | 25    | 50   |

<sup>1</sup> Per MISO BPM 015-r19 the following dispatch assumptions are applied for each Fuel Type.

- a. Combined Cycle (CC) is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak models.
- b. Solar is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak models.
- c. Wind is dispatched to 100% of the Requested MW in the shoulder models, 15.6% in the summer peak models.
- d. Battery is dispatched, both in charging and discharging modes, to 100% of the Requested MW in all study models.

## **1.2 Generating Facility Requirements**

### **1.2.1 Voltage Schedule Requirement**

ATC requires all generators in its territory to maintain a voltage schedule at the Point of Interconnection (POI). The standard voltage schedule is 1.02 per unit as measured at the POI. This schedule may be changed by the Transmission Operator for specific power plants or specific conditions.

### **1.2.2 Power Factor Range Requirement**

FERC Order 827 and ATC Criteria require all newly interconnecting generators interconnecting to ATC-owned Facilities to provide a power factor range for synchronous and non-synchronous (e.g., wind turbines, solar) generation of 0.95 leading (when a Generating Facility is consuming reactive power from the Transmission System) to 0.95 lagging (when a Generating Facility is supplying reactive power to the Transmission System). The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing dynamic reactive power at the following locations:

- A. The POI for all synchronous generators
- B. The high-side of the generator substation for all non-synchronous generators

For synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI. For non-synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the generator substation. Dynamic reactive power provided by non-synchronous generators must meet the following requirement from FERC order 827 Item 35:

"Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses."

Therefore, static reactive power sources can only be used to make up for losses between the terminal of the machines and the high side of the generator substation for non-synchronous machines. All other reactive power to meet the power factor requirement must be provided by dynamic sources. Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Capacitor Banks are considered static sources of reactive power. Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters and synchronous condensers are all considered dynamic sources of reactive power.

For non-synchronous generation projects in the Apr 18 DPP Phase 1 study group, if they did not have a signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The generation requests shown in Table 1.2.2-1 did not meet the dynamic reactive power requirements per FERC Order 827 and ATC Criteria at the time of the model completion and are required to provide additional dynamic reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for dynamic reactive power requirements.

**Table 1.2.2-1 – Additional Dynamic Mvar to meet ATC Dynamic Inductive Power Factor Requirement and FERC Order 827 Power Factor Requirements**

| <b>MISO Queue #</b> | <b>Type</b>  | <b>Additional Dynamic Reactive Compensation (Mvar)</b> |
|---------------------|--------------|--------------------------------------------------------|
| J1171               | Asynchronous | -1.90                                                  |
| J1188               | Asynchronous | -0.92                                                  |

The generation requests shown in Table 1.2.2-2 did not meet the static reactive power requirements per ATC Criteria at the time of the model completion and are required to provide additional static reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for static reactive power requirements.

**Table 1.2.2-2 – Additional Static Mvar to meet ATC Capacitive Power Factor and FERC Order 827 Power Factor Requirements**

| <b>MISO Queue #</b> | <b>Type</b>  | <b>Additional Static Shunt Compensation <sup>1</sup> (Mvar)</b> |
|---------------------|--------------|-----------------------------------------------------------------|
| J1009               | Asynchronous | 35.6                                                            |
| J1010               | Asynchronous | 38.3                                                            |
| J1011               | Asynchronous | 35.6                                                            |
| J1042               | Asynchronous | 19.3                                                            |
| J1044               | Asynchronous | 19.3                                                            |
| J1047               | Asynchronous | 19.3                                                            |
| J1101               | Asynchronous | 0.7                                                             |
| J1121               | Asynchronous | 16.3                                                            |
| J1127               | Asynchronous | 11.0                                                            |
| J1129               | Asynchronous | 12.7                                                            |
| J1153               | Asynchronous | 8.4                                                             |
| J1171               | Asynchronous | 13.9                                                            |
| J1188               | Asynchronous | 10.1                                                            |

<sup>1</sup> Additional compensation is required to meet the criteria at the POI Bus for synchronous Generating Facilities or the high-side of the generator substation for asynchronous Generating Facilities.

The generation requests shown in Table 1.2.2-3 may need an assessment of power factor at low real power output levels. More information regarding reactive power capability under low real power output level is required from the interconnection customers to determine whether the study is needed. All other requests in this queue met FERC Order 827 and ATC Criteria and do not need an assessment of power factor requirements at low real power output levels.

**Table 1.2.2-3 – Assessment of Power Factor Requirements at Low Output Levels**

| MISO Queue # | P-Q Curve Type           | Is Power Factor Evaluation Needed for Minimum Output Levels? | Additional Static Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Capacitive Power Factor Requirement |
|--------------|--------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| J1085        | D-shape with restriction | Yes                                                          | TBD                                                                                                                               |
| J1121        | D-shape with restriction | Yes                                                          | TBD                                                                                                                               |

### 1.2.3 Island Detection and Operation

In circumstances where the Generating Facility has no governor controls and the transmission system design could result in an islanding condition for the outage of two transmission elements, ATC requires the Customer to implement additional protection systems as mutually agreed by the Customer and ATC to prevent generation from being isolated or islanded with interconnected load. Alternatively, ATC will require the Customer to curtail their generation for circumstances that could result in an island condition with the next contingency.

This would apply to the following Generating Facilities from this DPP cycle that lack adequate governor controls to safely and reliably sustain an island with load.

- J1104
- J1127
- J1153
- J1171

### 1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service and Network Resource Interconnection Service as of the SIS report date. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1.3-1. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies. No Network Upgrade projects driven by stability, short circuit and PJM affected system analysis were not identified since these analyses were not performed in Phase 1. All Interconnection Facility Project Diagrams are documented in Appendix C and all Network Upgrade Project

Diagrams are documented in Appendix D (No project diagrams are developed for line uprates).

Table 1.3-1 – Total Cost of Network Upgrades for April 18 DPP Phase 1 Generator Interconnection Requests

| MISO Queue # | Requested MW | ERIS Network Upgrades (\$) <sup>7</sup> |                     |               |                 | NRIS Network Upgrades (\$) | Interconnection Facilities (\$) |                                 | Shared Network Upgrade (\$) | Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$) | M2 Received (\$)¹  | M3 Due (\$)²         |
|--------------|--------------|-----------------------------------------|---------------------|---------------|-----------------|----------------------------|---------------------------------|---------------------------------|-----------------------------|-----------------------------------------------------------------------|--------------------|----------------------|
|              |              | Steady - State Thermal & Voltage        | Transient Stability | Short Circuit | Affected System |                            | TO Network Upgrades             | TO-Owned Direct Assigned (TOIF) |                             |                                                                       |                    |                      |
| [a]          | [b]          | [c]                                     | [d]                 | [e]           | [f]             | [g]                        | [h]                             | [i]                             | [j]                         | [k]=[c]+[d]+[e]+[g]+[h]                                               | [l] = \$4000 x [b] | [m]= (10%of [k])-[l] |
| J986         | 149.76       | 5,227,784                               | 0                   | 0             | 0               | 0                          | 8,529,512                       | 335,442                         | 0                           | 13,757,296                                                            | 599,040            | 776,690              |
| J1000        | 50           | 0                                       | 0                   | 0             | 0               | 0                          | 1,043,703                       | 335,825                         | 0                           | 1,043,703                                                             | 200,000            | 0                    |
| J1002        | 99           | 1,531,777                               | 0                   | 0             | 0               | 0                          | 2,015,000                       | 315,292                         | 0                           | 3,546,777                                                             | 396,000            | 0                    |
| J1003        | 50           | 341,097                                 | 0                   | 0             | 0               | 0                          | 1,040,867                       | 166,604                         | 0                           | 1,381,964                                                             | 200,000            | 0                    |
| J1009        | 400          | 0                                       | 0                   | 0             | 0               | 0                          | 13,350,471                      | 638,145                         | 0                           | 13,350,471                                                            | 1,600,000          | 0                    |
| J1010        | 400          | 0                                       | 0                   | 0             | 0               | 827,016                    | 4,376,063                       | 267,432                         | 0                           | 5,203,079                                                             | 1,600,000          | 0                    |
| J1011        | 400          | 0                                       | 0                   | 0             | 0               | 827,016                    | 4,376,063                       | 267,432                         | 0                           | 5,203,079                                                             | 1,600,000          | 0                    |
| J1042        | 200          | 2,418,767                               | 0                   | 0             | 0               | 7,726,710                  | 444,446                         | 365,492                         | 0                           | 10,589,923                                                            | 800,000            | 258,992              |
| J1044        | 200          | 3,938,539                               | 0                   | 0             | 0               | 9,759,486                  | 4,779,657                       | 209,065                         | 0                           | 18,477,682                                                            | 800,000            | 1,047,768            |
| J1047        | 200          | 3,938,539                               | 0                   | 0             | 0               | 9,759,486                  | 4,779,657                       | 209,065                         | 0                           | 18,477,682                                                            | 800,000            | 1,047,768            |
| J1051        | 50           | 5,392,870                               | 0                   | 0             | 250,000         | 0                          | 5,486,526                       | 129,664                         | 0                           | 10,879,396                                                            | 200,000            | 887,940              |
| J1053        | 300          | 26,207,430                              | 0                   | 0             | 250,000         | 0                          | 5,486,526                       | 129,664                         | 0                           | 31,693,956                                                            | 1,200,000          | 1,969,396            |
| J1085        | 300          | 8,841,697                               | 0                   | 0             | 0               | 0                          | 8,631,561                       | 331,811                         | 0                           | 17,473,258                                                            | 1,200,000          | 547,326              |
| J1101        | 20           | 25,023                                  | 0                   | 0             | 0               | 0                          | 0                               | 0                               | 0                           | 25,023                                                                | 80,000             | 0                    |
| J1104        | 100          | 0                                       | 0                   | 0             | 0               | 6,271,958                  | 7,728,822                       | 262,301                         | 0                           | 14,000,780                                                            | 400,000            | 1,000,078            |
| J1121        | 200          | 4,578,540                               | 0                   | 0             | 0               | 0                          | 8,631,561                       | 331,811                         | 0                           | 13,210,101                                                            | 800,000            | 521,010              |
| J1127        | 49.9         | 127,717                                 | 0                   | 0             | 0               | 2,750,371                  | 7,287,114                       | 261,018                         | 0                           | 10,165,202                                                            | 199,600            | 816,920              |
| J1129        | 65           | 162,919                                 | 0                   | 0             | 0               | 1,285,314                  | 7,158,859                       | 261,018                         | 0                           | 8,607,091                                                             | 260,000            | 600,709              |
| J1153        | 150          | 0                                       | 0                   | 0             | 0               | 0                          | 1,465,758                       | 793,885                         | 0                           | 1,465,758                                                             | 600,000            | 0                    |
| J1154        | 75           | 1,877,380                               | 0                   | 0             | 0               | 0                          | 0                               | 0                               | 0                           | 1,877,380                                                             | 300,000            | 0                    |
| J1171        | 100          | 0                                       | 0                   | 0             | 0               | 0                          | 1,166,890                       | 320,252                         | 0                           | 1,166,890                                                             | 400,000            | 0                    |
| J1183        | 1.35         | 0                                       | 0                   | 0             | 0               | 0                          | 0                               | 0                               | 0                           | 0                                                                     | 5,400              | 0                    |
| J1188        | 50           | 106,020                                 | 0                   | 0             | 0               | 1,875,158                  | 1,377,714                       | 109,542                         | 0                           | 3,358,892                                                             | 200,000            | 135,889              |
| Total (\$)   | -            | 64,716,100                              | 0                   | 0             | 500,000         | 41,082,515                 | 99,156,770                      | 6,040,758                       | 0                           | 204,955,385                                                           | 14,440,040         | 9,610,486            |

<sup>1</sup> M2: Milestone Payment dollars received by MISO

<sup>2</sup> M3 = (10% of NU)-M2

<sup>3</sup> ATC projects are estimated in ISD dollars with a 2.5% annual escalation rate and include a 10% contingency.

<sup>4</sup> Pre-Certification costs are not included.

<sup>5</sup> Transmission Owner shall also collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer. The current Transmission Owner tax gross-up rate is 12.848%.

<sup>6</sup> TOIF: Non-reimbursable Transmission Owner Interconnection Facilities to which tax gross-up amount must also be applied.

<sup>7</sup> Transient stability, short circuit, and affected system studies will be performed as part of MISO DPP April 2018 Phase 2.

<sup>8</sup> Cost is reflective of changing the point of change of ownership (PCO) for We Energies CTs.

## 1.4 In-Service Dates and Cost Estimates

ATC understands that the estimated in-service date may not align with the Interconnection Customer's Synchronization Date; however, negotiated and executed agreements, such as an Engineering and Procurement Agreement, can be used prior to the GIA execution date to expedite Network Upgrades. In absence of any special arrangement, typical times to develop a new Interconnection Facility is about 24-36 months after the GIA is executed, assuming no delays due to Interconnection Customer's permits, state processes, land acquisitions, deliverables (such as a finish graded substation site, etc.) It also assumes that system outages required to construct facilities can be obtained timely. The cost estimates for Interconnection Facilities are based on the in-service date provided in the Interconnection Customer's application data. Therefore, any change in in-service date will have impact on the cost estimates. The requested dates for Interconnection Facility in-service, synchronization, and commercial operation are summarized in Table 1.4-1.

**Table 1.4-1 – Requested Interconnection Facilities In-Service Dates, Synchronization Dates and Commercial Operation Dates**

| MISO Queue # | Requested Interconnection Facility In-service Date | Requested Synchronization Date | Requested Commercial Operation Date |
|--------------|----------------------------------------------------|--------------------------------|-------------------------------------|
| J986         | September 1, 2020                                  | September 15, 2020             | December 1, 2020                    |
| J1000        | October 1, 2020                                    | October 15, 2020               | December 1, 2020                    |
| J1002        | October 1, 2021                                    | October 15, 2021               | December 1, 2021                    |
| J1003        | October 1, 2021                                    | October 15, 2021               | December 1, 2021                    |
| J1009        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1010        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1011        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1042        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1044        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1047        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1051        | October 31, 2021                                   | October 1, 2021                | December 31, 2021                   |
| J1053        | October 31, 2021                                   | October 1, 2021                | December 31, 2021                   |
| J1085        | September 1, 2021                                  | October 1, 2021                | December 1, 2021                    |
| J1101        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1104        | September 1, 2020                                  | October 1, 2020                | December 1, 2020                    |
| J1121        | September 1, 2021                                  | October 1, 2021                | December 1, 2021                    |
| J1127        | January 2, 2021                                    | February 1, 2021               | June 1, 2021                        |
| J1129        | January 2, 2021                                    | February 1, 2021               | June 1, 2021                        |
| J1153        | April 1, 2021                                      | April 15, 2021                 | June 30, 2021                       |
| J1154        | April 1, 2021                                      | April 15, 2021                 | June 30, 2021                       |

|       |                  |                    |                   |
|-------|------------------|--------------------|-------------------|
| J1171 | August 1, 2020   | October 1, 2020    | December 1, 2020  |
| J1183 | January 31, 2019 | January 31, 2019   | January 31, 2019  |
| J1188 | August 31, 2020  | September 30, 2020 | November 30, 2020 |

## 1.5 MTEP Projects

If a MTEP transmission project(s) resolves the constraint, and that project(s) is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the Interconnection Customer will not be responsible for transmission upgrade(s) that would resolve the constraint. If that MTEP project(s) is not approved within one (1) calendar year of the GIA execution or execution of an amendment thereof, the Interconnection Customer will be responsible for those transmission upgrade(s).

## 1.6 Further Study

The next step in the MISO Generator Interconnection Procedures is to perform additional SISs (if needed), Interconnection Customer Interconnection Facility Studies, and Network Upgrade Facility Studies. Those Facilities Studies will specify in more detail the time and cost of the equipment, engineering, procurement, and construction of the Interconnection Facilities and Network Upgrades identified in this report.

## 1.7 Compliance Summary

This study report partially meets NERC TPL-001-4 standard, FAC-002-2 standard, and Local Planning Criteria. In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the generator interconnection requests with signed GIAs. Appendix J describes in detail the NERC and Local Criteria requirements met by this SIS report.

## 2.0 STEADY-STATE ANALYSIS

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generator interconnection requests in the Apr 18 DPP Phase 1 to the transmission system. Detailed study assumptions, criteria, and methodology are documented in Appendix A.

### 2.1 Model Development

#### 2.1.1 Study Cases

Four study cases for the steady-state thermal and voltage analysis were developed based upon the expected topology for the local area: 2023 summer peak batteries discharging, 2023 summer peak batteries charging, 2023 shoulder batteries discharging, and 2023 shoulder batteries charging. The ATC system in these cases was updated with the most recent data available at the time of model construction. The Cardinal – Hickory Creek project was included in both

study models, even though its in-service date is 12/31/2023, because it was defined as a required Network Upgrade in the DPP 2017 February Wisconsin Area Phase 1 SIS. The cases were reviewed by ATC and the Interconnection Customers. Based on this review, the cases were further modified to account for model updates, changes, and competing generation requests that had dropped out of the MISO queue since the models were built.

The prior queued generator interconnection requests in the ATC system that are included in the study cases are listed in Table 2.1.1-1. Associated Network Upgrades were also included based on their expected in-service date.

**Table 2.1.1-1 – Prior Queued Generator Interconnection Requests  
Not Yet In-Service**

| MISO Queue # | Type  | Control Area | Requested MW | Requested In-Service Year |
|--------------|-------|--------------|--------------|---------------------------|
| J390         | CC    | ALTE         | 702          | 2019                      |
| J505         | Solar | WPS          | 99           | 2021                      |
| J584         | Wind  | ALTE         | 60           | 2019                      |
| J732         | CC    | WPS          | 561.5        | 2021                      |
| J760         | CC    | ALTE         | 30           | 2019                      |
| J807         | Wind  | ALTE         | 41.4         | 2020                      |
| J818         | Solar | WEC          | 149          | 2019                      |
| J819         | Wind  | ALTE         | 99.9         | 2020                      |
| J821         | Wind  | WPS          | 99.9         | 2020                      |
| J831         | CC    | WEC          | 48           | 2017                      |
| J849         | Solar | MIUP         | 125          | 2020                      |
| J850         | Solar | ALTE         | 250          | 2019                      |
| J855         | Wind  | ALTE         | 100          | 2021                      |
| J864         | Solar | ALTE         | 49.98        | 2019                      |
| J870         | Solar | ALTE         | 200          | 2021                      |
| J871         | Solar | ALTE         | 100          | 2021                      |
| J878         | Solar | WEC          | 200          | 2021                      |
| J886         | Solar | WPS          | 150          | 2020                      |
| J928         | Wind  | MIUP         | 79.995       | 2019                      |
| J947         | Solar | ALTE         | 200          | 2019                      |

Public information related to the MISO Generator Interconnection Request queue can be found at:

[https://www.misoenergy.org/planning/generator-interconnection/GI\\_Queue/](https://www.misoenergy.org/planning/generator-interconnection/GI_Queue/)

The summer peak discharging, summer peak charging, shoulder discharging, and shoulder charging cases dispatched generation within MISO according to section 6.1.1.1.2, Study Case Development, in the MISO BPM-015-r19.

All excess generation from this methodology is dispatched against all units in MISO Classic proportionally, excluding the units in the current DPP cycle. Scheduled firm transfers are ignored in this dispatch methodology.

## 2.1.2 Benchmark Cases

Two benchmark cases were used to benchmark system performance without the Apr 18 DPP Phase 1 generating facilities and were created by taking the Apr 18

DPP Phase 1 Generating Facilities offline from the corresponding study cases. The MISO Classic was used for power balance, where generation was scaled in proportion to  $P_{\max}$  minus  $P_{\text{gen}}$ .

## **2.2 Reactive Power Requirements (FERC Order 827)**

All synchronous and non-synchronous generation in this queue were evaluated to determine if the requests meet FERC Order 827 and ATC Planning Criteria. Refer to PLG-METH-0005 in Appendix B for details on the methodology used to determine power factor compliance. All of the reactive resources modeled in the assessment are summarized in Table 2.2-1.

Table 2.2-1 – Reactive Resources Modeled in Generator Interconnection Power Factor Analysis

| MISO Queue # | Machines                        |                 |                                  |                                 | Dynamic Devices |                                  |                                 | Static Devices               |                                  |                                 |
|--------------|---------------------------------|-----------------|----------------------------------|---------------------------------|-----------------|----------------------------------|---------------------------------|------------------------------|----------------------------------|---------------------------------|
|              | Description                     | Real Power (MW) | Capacitive Reactive Power (Mvar) | Inductive Reactive Power (Mvar) | Description     | Capacitive Reactive Power (Mvar) | Inductive Reactive Power (Mvar) | Description                  | Capacitive Reactive Power (Mvar) | Inductive Reactive Power (Mvar) |
| J986         | Solar Machine                   | 149.76          | 72.49                            | -72.49                          | None            | N/A                              | N/A                             | 6 * 4.5 Mvar Capacitor Banks | 27                               | 0                               |
| J1000        | Solar Machine                   | 50              | 29.668                           | -29.668                         | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1002        | Solar Machine                   | 99              | 59                               | -59                             | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1003        | Solar Machine                   | 50              | 29.668                           | -29.668                         | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1009        | Solar Machine                   | 400             | 163.2                            | -163.2                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1010        | Solar Machine                   | 400             | 163.2                            | -163.2                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1011        | Solar Machine                   | 400             | 163.2                            | -163.2                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1042        | Solar Machine                   | 200             | 81.6                             | -81.6                           | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1044        | Solar Machine                   | 200             | 81.6                             | -81.6                           | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1047        | Solar Machine                   | 200             | 81.6                             | -81.6                           | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1051        | Battery Storage                 | 51.0            | 19.49                            | -19.49                          | None            | N/A                              | N/A                             | 20 Mvar Capacitor Bank       | 20                               | 0                               |
| J1053        | Solar Machine                   | 306             | 138.8                            | -138.8                          | None            | N/A                              | N/A                             | 2 * 42 Mvar Capacitor Banks  | 84                               | 0                               |
| J1085        | Wind Turbines                   | 300             | 145.3                            | -145.3                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1101        | Battery Storage                 | 20              | 6.574                            | -6.574                          | None            | N/A                              | N/A                             | 3 Mvar Capacitor Bank        | 3                                | 0                               |
| J1104        | Battery Storage                 | 100             | 32.36                            | -32.36                          | None            | N/A                              | N/A                             | 2 * 15 Mvar Capacitor Banks  | 30                               | 0                               |
| J1121        | Solar Machine                   | 200             | 87.6                             | -87.6                           | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1127        | Solar Machine                   | 49.9            | 16.401                           | -16.401                         | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1129        | Solar Machine                   | 65.00           | 21.364                           | -21.364                         | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1153        | Solar Machine                   | 150             | 68.7386                          | -68.7386                        | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1154        | Solar Machine                   | 75              | 34.3693                          | -34.3693                        | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1171        | Solar Machine                   | 102.58          | 33.92                            | -31.82                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |
| J1183        | Solar Machines and Wind Machine | 30.25           | 10.1005                          | -10.005                         | None            | N/A                              | N/A                             | 10.5 Mvar Capacitor Bank     | 10.5                             | 0                               |
| J1188        | Solar Machine                   | 51.29           | 16.960                           | -15.91                          | None            | N/A                              | N/A                             | None                         | 0                                | 0                               |

The dynamic capacitive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-2.

**Table 2.2-2 – Assessment of Dynamic Capacitive Power Factor Requirement**

| MISO Queue # | Machine Terminal Bus # | Capability at Machine Terminal |                                  | Additional Dynamic Capacitive Reactive Power <sup>1</sup> (Mvar) | Dynamic Power Factor Provided | Meets Requirement <sup>2</sup> ? | Additional Requirement <sup>3</sup> (Mvar) |
|--------------|------------------------|--------------------------------|----------------------------------|------------------------------------------------------------------|-------------------------------|----------------------------------|--------------------------------------------|
|              |                        | Real Power (MW)                | Capacitive Reactive Power (Mvar) |                                                                  |                               |                                  |                                            |
| J986         | 89860                  | 149.8                          | 72.490                           | 0.0                                                              | 0.90                          | Yes                              | 0.0                                        |
| J1000        | 40000                  | 50.0                           | 29.668                           | 0.0                                                              | 0.86                          | Yes                              | 0.0                                        |
| J1002        | 40020                  | 99.0                           | 58.743                           | 0.0                                                              | 0.86                          | Yes                              | 0.0                                        |
| J1003        | 40030                  | 50.00                          | 29.700                           | 0.0                                                              | 0.86                          | Yes                              | 0.0                                        |
| J1009        | 40090                  | 400.0                          | 163.2                            | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1010        | 40100                  | 400.0                          | 163.2                            | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1011        | 40110                  | 400.0                          | 163.2                            | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1042        | 40420                  | 200.0                          | 81.6                             | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1044        | 40440                  | 200.0                          | 81.6                             | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1047        | 40470                  | 200.0                          | 81.6                             | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1051        | 40510                  | 51.0                           | 19.49                            | 0.0                                                              | 0.93                          | Yes                              | 0.0                                        |
| J1053        | 40530                  | 306                            | 138.8                            | 0.0                                                              | 0.91                          | Yes                              | 0.0                                        |
| J1085        | 40850, 40858           | 300.0                          | 145.300                          | 0.0                                                              | 0.90                          | Yes                              | 0.0                                        |
| J1101        | 41010                  | 20.000                         | 6.574                            | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1104        | 40040                  | 100                            | 32.36                            | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1121        | 41210                  | 200.0                          | 87.600                           | 0.0                                                              | 0.92                          | Yes                              | 0.0                                        |
| J1127        | 41270                  | 49.9                           | 16.401                           | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1129        | 41290                  | 65.00                          | 21.364                           | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1153        | 41530                  | 150.0                          | 68.739                           | 0.0                                                              | 0.91                          | Yes                              | 0.0                                        |
| J1154        | 40540                  | 75                             | 34.3693                          | 0.0                                                              | 0.91                          | Yes                              | 0.0                                        |
| J1171        | 40710                  | 102.58                         | 33.92                            | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1183        | 41830, 693697, 693704  | 30.25                          | 10.1005                          | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |
| J1188        | 41880                  | 51.29                          | 16.960                           | 0.0                                                              | 0.95                          | Yes                              | 0.0                                        |

<sup>1</sup> Dynamic capacitive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

<sup>2</sup> ATC requires a 0.95 ATC Capacitive Dynamic Power Factor.

<sup>3</sup> Additional dynamic reactive power required to meet ATC Capacitive Dynamic Power Factor.

The static capacitive power factor requirement analysis showed 13 of the 23 requests do not meet the ATC Criteria or FERC Order 827 requirements. The results are summarized in Table 2.2-3.

**Table 2.2-3 – Assessment of Static Capacitive Power Factor Requirement**

| MISO Queue # | Point of Measurement | Capability at Point of Measurement <sup>1</sup> |                       | Power Factor | Meets Requirement? | Additional Requirement (Mvar) |
|--------------|----------------------|-------------------------------------------------|-----------------------|--------------|--------------------|-------------------------------|
|              |                      | Real Power (MW)                                 | Reactive Power (Mvar) |              |                    |                               |
| J986         | 89863                | 146.2                                           | 67                    | 0.91         | Yes                | 0.0                           |
| J1000        | 40003                | 49.2                                            | 19.9                  | 0.93         | Yes                | 0.0                           |
| J1002        | 40023                | 97                                              | 37                    | 0.93         | Yes                | 0.0                           |
| J1003        | 40033                | 49                                              | 20.2                  | 0.92         | Yes                | 0.0                           |
| J1009        | 40090                | 394.4                                           | 95.2                  | 0.97         | No                 | 34.4                          |
| J1010        | 40100                | 394.4                                           | 95.2                  | 0.97         | No                 | 34.4                          |
| J1011        | 40110                | 394.4                                           | 95.2                  | 0.97         | No                 | 34.4                          |
| J1042        | 40420                | 197.0                                           | 45.6                  | 0.97         | No                 | 17.1                          |
| J1044        | 40440                | 197.0                                           | 45.6                  | 0.97         | No                 | 17.1                          |
| J1047        | 40470                | 197.0                                           | 45.6                  | 0.97         | No                 | 17.1                          |
| J1051        | 990001               | 50                                              | 33.5                  | 0.83         | Yes                | 0.0                           |
| J1053        | 990001               | 299.8                                           | 141.8                 | 0.90         | Yes                | 0.0                           |
| J1085        | 40854                | 290.1                                           | 101.9                 | 0.94         | Yes                | 0.0                           |
| J1101        | 88604                | 19.8                                            | 5.8                   | 0.96         | No                 | 0.7                           |
| J1104        | 41043                | 98.6                                            | 42.1                  | 0.92         | Yes                | 0.0                           |
| J1121        | 40854                | 195.3                                           | 47.9                  | 0.97         | No                 | 16.3                          |
| J1127        | 41273                | 49.2                                            | 5.2                   | 0.99         | No                 | 11.0                          |
| J1129        | 41293                | 63.9                                            | 8.3                   | 0.99         | No                 | 12.7                          |
| J1153        | 41533                | 147.8                                           | 40.2                  | 0.96         | No                 | 8.4                           |
| J1154        | 41543                | 74                                              | 25.7                  | 0.94         | Yes                | 0.0                           |
| J1171        | 41713                | 100                                             | 19                    | 0.98         | No                 | 13.9                          |
| J1183        | 693702               | 29.7                                            | 14.3                  | 0.90         | Yes                | 0.0                           |
| J1188        | 41883                | 50.6                                            | 6.5                   | 0.99         | No                 | 10.1                          |

<sup>1</sup> Point of Measurement is the POI Bus for synchronous machines and high side of generator substation for asynchronous machines.

The dynamic inductive power factor requirement analysis showed all but two of the requests meet ATC Criteria and FERC Order 827 requirements. J1171 and J1188 do not meet the ATC Criteria and FERC Order 827 requirements and must provide additional dynamic resources prior to Phase two. The results are summarized in Table 2.2-4.

**Table 2.2-4 – Assessment of Dynamic Inductive Power Factor Requirement**

| MISO Queue # | Machine Terminal Bus # | Capability at Machine Terminal |                                 | Additional Dynamic Inductive Reactive Power <sup>1</sup> (Mvar) | Dynamic Power Factor Provided | Meets Requirement <sup>2</sup> ? | Additional Requirement <sup>3</sup> (Mvar) |
|--------------|------------------------|--------------------------------|---------------------------------|-----------------------------------------------------------------|-------------------------------|----------------------------------|--------------------------------------------|
|              |                        | Real Power (MW)                | Inductive Reactive Power (Mvar) |                                                                 |                               |                                  |                                            |
| J986         | 89860                  | 149.8                          | -72.49                          | 0.0                                                             | 0.90                          | Yes                              | 0.0                                        |
| J1000        | 40000                  | 50.0                           | -29.668                         | 0.0                                                             | 0.86                          | Yes                              | 0.0                                        |
| J1002        | 40020                  | 99.0                           | -58.7431                        | 0.0                                                             | 0.86                          | Yes                              | 0.0                                        |
| J1003        | 40030                  | 50.00                          | -29.7                           | 0.0                                                             | 0.86                          | Yes                              | 0.0                                        |
| J1009        | 40090                  | 400.0                          | -163.2                          | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1010        | 40100                  | 400.0                          | -163.2                          | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1011        | 40110                  | 400.0                          | -163.2                          | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1042        | 40420                  | 200.0                          | -81.6                           | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1044        | 40440                  | 200.0                          | -81.6                           | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1047        | 40470                  | 200.0                          | -81.6                           | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1051        | 40510                  | 200                            | -81.6                           | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1053        | 40530                  | 51.0                           | -19.49                          | 0.0                                                             | 0.93                          | Yes                              | 0.0                                        |
| J1085        | 40850, 40858           | 300.0                          | -145.300                        | 0.0                                                             | 0.90                          | Yes                              | 0.0                                        |
| J1101        | 41010                  | 20.000                         | -6.574                          | 0.0                                                             | 0.95                          | Yes                              | 0.0                                        |
| J1104        | 40400                  | 100.0                          | -32.36                          | 0.0                                                             | 0.95                          | Yes                              | 0.0                                        |
| J1121        | 41210                  | 200.0                          | -87.600                         | 0.0                                                             | 0.92                          | Yes                              | 0.0                                        |
| J1127        | 41270                  | 49.9                           | -16.401                         | 0.0                                                             | 0.95                          | Yes                              | 0.0                                        |
| J1129        | 41290                  | 65.00                          | -21.364                         | 0.0                                                             | 0.95                          | Yes                              | 0.0                                        |
| J1153        | 41530                  | 150.0                          | -68.7386                        | 0.0                                                             | 0.91                          | Yes                              | 0.0                                        |
| J1154        | 40540                  | 75                             | -34.3693                        | 0.0                                                             | 0.91                          | Yes                              | 0.0                                        |
| J1171        | 40710                  | 102.58                         | -31.82                          | 0.0                                                             | 0.96                          | No                               | -1.90                                      |
| J1183        | 41830, 693697, 693704  | 30.25                          | -10.1005                        | 0.0                                                             | 0.95                          | Yes                              | 0.0                                        |
| J1188        | 41880                  | 51.29                          | -15.91                          | 0.0                                                             | 0.96                          | No                               | -0.92                                      |

<sup>1</sup> Dynamic inductive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

<sup>2</sup> ATC requires a 0.95 ATC Inductive Dynamic Power Factor.

<sup>3</sup> Additional dynamic reactive power required to meet ATC Inductive Dynamic Power Factor.

The static inductive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-5.

**Table 2.2-5 – Assessment of Static Inductive Power Factor Requirement**

| MISO Queue # | POI Bus (synchronous) or HV Bus (asynchronous) | MW at POI (synchronous) or HV Bus (asynchronous) | Mvar at POI (synchronous) or HV Bus (asynchronous) | POI (synchronous) or HV Bus (asynchronous) Power Factor | Meets 0.95 ATC Inductive Power Factor Requirement at POI Bus (synchronous) or HV Bus (asynchronous)? |
|--------------|------------------------------------------------|--------------------------------------------------|----------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| J986         | 89863                                          | 145.8                                            | -109.9                                             | 0.80                                                    | Yes                                                                                                  |
| J1000        | 40003                                          | 49.0                                             | -42.9                                              | 0.75                                                    | Yes                                                                                                  |
| J1002        | 40023                                          | 96.4                                             | -88.4                                              | 0.74                                                    | Yes                                                                                                  |
| J1003        | 40033                                          | 48.7                                             | -42.6                                              | 0.75                                                    | Yes                                                                                                  |
| J1009        | 40090                                          | 393.2                                            | -262.0                                             | 0.83                                                    | Yes                                                                                                  |
| J1010        | 40100                                          | 393.2                                            | -262.0                                             | 0.83                                                    | Yes                                                                                                  |
| J1011        | 40110                                          | 393.2                                            | -262.0                                             | 0.83                                                    | Yes                                                                                                  |
| J1042        | 40420                                          | 196.5                                            | -128.3                                             | 0.84                                                    | Yes                                                                                                  |
| J1044        | 40440                                          | 196.4                                            | -130.9                                             | 0.83                                                    | Yes                                                                                                  |
| J1047        | 40470                                          | 196.5                                            | -128.3                                             | 0.84                                                    | Yes                                                                                                  |
| J1051        | 990001                                         | 50.0                                             | -23.6                                              | 0.90                                                    | Yes                                                                                                  |
| J1053        | 990001                                         | 297.6                                            | 244.6                                              | 0.77                                                    | Yes                                                                                                  |
| J1085        | 40854                                          | 289.2                                            | 193.9                                              | 0.83                                                    | Yes                                                                                                  |
| J1101        | 886004                                         | 19.7                                             | -10.2                                              | 0.89                                                    | Yes                                                                                                  |
| J1104        | 41043                                          | 98.5                                             | -52.5                                              | 0.88                                                    | Yes                                                                                                  |
| J1121        | 40854                                          | 195.0                                            | -132.4                                             | 0.83                                                    | Yes                                                                                                  |
| J1127        | 41273                                          | 49.1                                             | -28.5                                              | 0.86                                                    | Yes                                                                                                  |
| J1129        | 41293                                          | 63.9                                             | -35.1                                              | 0.88                                                    | Yes                                                                                                  |
| J1153        | 41533                                          | 147.1                                            | -106.3                                             | 0.81                                                    | Yes                                                                                                  |
| J1154        | 41543                                          | 74.0                                             | -42.8                                              | 0.87                                                    | Yes                                                                                                  |
| J1171        | 41713                                          | 100.0                                            | -46.1                                              | 0.91                                                    | Yes                                                                                                  |
| J1183        | 693702                                         | 29.6                                             | -15.4                                              | 0.89                                                    | Yes                                                                                                  |
| J1188        | 41883                                          | 50.6                                             | -26.7                                              | 0.88                                                    | Yes                                                                                                  |

The low output power factor requirement analysis showed all requests except J1085 and J1121 meet ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-6.

**Table 2.2-6 – Assessment of Power Factor Requirements at Low Output Levels**

| MISO Queue # | P-Q Curve Type           | Is Power Factor Evaluation Needed for Minimum Output Levels? | Additional Static Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Capacitive Power Factor Requirement |
|--------------|--------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| J986         | U-Shape                  | yes                                                          | no                                                                                                                                |
| J1000        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1002        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1003        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1009        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1010        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1011        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1042        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1044        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1047        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1051        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1053        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1085        | D-shape with restriction | yes <sup>1</sup>                                             | TBD                                                                                                                               |
| J1101        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1104        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1121        | D-shape with restriction | yes <sup>1</sup>                                             | TBD                                                                                                                               |
| J1127        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1129        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1153        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1154        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1171        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1183        | D-shape                  | no                                                           | N/A                                                                                                                               |
| J1188        | D-shape                  | no                                                           | N/A                                                                                                                               |

<sup>1</sup>More information regarding reactive power capability under low real power output level is required from interconnection customers to perform this study.

## 2.3 NERC TPL Contingency Analysis Results

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases (without Apr 18 DPP Phase 1 projects) and study

cases (with Apr 18 DPP Phase 1 projects). Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. Detailed NERC TPL Category P contingencies that were studied are described in Table A.2.1-1 in Appendix A.

### **2.3.1 2023 Summer Peak Batteries Discharging**

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Summer Peak Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Summer Peak Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.1-1. No voltage constraints were identified in the steady-state analyses.

**Table 2.3.1 – 2023 Summer Peak Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades**

| Study Case              | Overloaded Facility                            | Facility Owner | Resulted by "No Load Loss Allowed" <sup>1</sup> NERC TPL Planning Events | Responsible Generator(s)                           |
|-------------------------|------------------------------------------------|----------------|--------------------------------------------------------------------------|----------------------------------------------------|
| 2023 Summer Discharging | Burlington – North Lake Geneva Tap 138 kV line | ATC            | Base Case<br>P11<br>P12<br>P13<br>P21<br>P23                             | J1042<br>J1044<br>J1047<br>J1127<br>J1129<br>J1188 |
|                         | Sand Lake 138/69 kV transformer                | ATC            | P12                                                                      | J986                                               |
|                         | Wautoma 138/69 kV transformer                  | ATC            | Base Case<br>P12<br>P13<br>P21                                           | J986<br>J1002                                      |
|                         | J986 POI – Port Edwards 138 kV line            | ATC            | P11<br>P12<br>P13<br>P21<br>P23<br>P55                                   | J986<br>J1002                                      |
|                         | Concord – Rubicon 138 kV line                  | ATC            | P12                                                                      | J1051<br>J1053                                     |
|                         | Summit 138 kV bus tie 2-3                      | ATC            | P12                                                                      | J1051<br>J1053<br>J1154                            |
|                         | Cooney – Summit 138 kV line                    | ATC            | Base Case<br>P12                                                         | J1003<br>J1051<br>J1053<br>J1154                   |
|                         | Concord – Cooney 138 kV line                   | ATC            | Base Case<br>P12                                                         | J1003<br>J1051<br>J1053<br>J1154                   |
|                         | Stone Lake 345/161 kV transformer              | XEL            | P12<br>P23                                                               | J1085<br>J1121                                     |

<sup>1</sup> The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

### 2.3.2 2023 Summer Peak Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Summer Peak Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Summer Peak Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.1-2. No voltage constraints were identified in the steady-state analyses.

**Table 2.3.2 – 2023 Summer Peak Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades**

| Study Case           | Overloaded Facility                            | Facility Owner | Resulted by "No Load Loss Allowed" <sup>1</sup> NERC TPL Planning Events | Responsible Generator(s)                           |
|----------------------|------------------------------------------------|----------------|--------------------------------------------------------------------------|----------------------------------------------------|
| 2023 Summer Charging | Burlington – North Lake Geneva Tap 138 kV line | ATC            | Base Case<br>P11<br>P12<br>P13<br>P21<br>P23                             | J1042<br>J1044<br>J1047<br>J1127<br>J1129<br>J1188 |
|                      | Sand Lake 138/69 kV transformer                | ATC            | P12                                                                      | J986<br>J1002                                      |
|                      | Wautoma 138/69 kV transformer                  | ATC            | Base Case<br>P12<br>P13<br>P21                                           | J986<br>J1002                                      |
|                      | J986 POI – Port Edwards 138 kV line            | ATC            | P11<br>P12<br>P13<br>P21<br>P23                                          | J986<br>J1002                                      |
|                      | Cooney – Summit 138 kV line                    | ATC            | Base Case<br>P12                                                         | J1003<br>J1051<br>J1053<br>J1154                   |
|                      | Concord – Cooney 138 kV line                   | ATC            | Base Case<br>P12                                                         | J1003<br>J1051<br>J1053<br>J1154                   |
|                      | Stone Lake 345/161 kV transformer              | XEL            | P12<br>P23                                                               | J1085<br>J1121                                     |

<sup>1</sup> The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

### 2.3.3 2023 Shoulder Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Shoulder Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Shoulder Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2-3. No voltage constraints were identified in the steady state analyses.

**Table 2.3.3 – 2023 Shoulder Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades**

| Study Case                | Overloaded Facility                            | Facility Owner | Resulted by "No Load Loss Allowed" <sup>1</sup> NERC TPL Planning Events | Responsible Generator(s)                           |
|---------------------------|------------------------------------------------|----------------|--------------------------------------------------------------------------|----------------------------------------------------|
| 2023 Shoulder Discharging | Burlington – North Lake Geneva Tap 138 kV line | ATC            | Base Case<br>P12<br>P23                                                  | J1042<br>J1044<br>J1047<br>J1127<br>J1129<br>J1188 |
|                           | Saratoga 138/69 kV transformer                 | ATC            | P23                                                                      | J986                                               |
|                           | Rocky Run – Werner West 345 kV line            | ATC            | P12<br>P23                                                               | J986<br>J1085<br>J1121                             |
|                           | Lublin – Lublin Pump 69 kV line                | DPC            | P12<br>P23                                                               | J1085<br>J1121                                     |
|                           | Lublin Tap – Lublin Pump 69 kV line            | DPC            | P12<br>P23                                                               | J1085<br>J1121                                     |
|                           | Stone Lake 345/161 kV transformer              | XEL            | P12<br>P23                                                               | J1085<br>J1121                                     |

<sup>1</sup> The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

### 2.3.4 2023 Shoulder Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2023 Shoulder Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2023 Shoulder Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2-4. No voltage constraints were identified in the steady state analyses.

**Table 2.3.4 – 2023 Shoulder Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades**

| Study Case             | Overloaded Facility                            | Facility Owner | Resulted by "No Load Loss Allowed" <sup>1</sup> NERC TPL Planning Events | Responsible Generator(s)                           |
|------------------------|------------------------------------------------|----------------|--------------------------------------------------------------------------|----------------------------------------------------|
| 2023 Shoulder Charging | Burlington – North Lake Geneva Tap 138 kV line | ATC            | Base Case<br>P12<br>P23                                                  | J1042<br>J1044<br>J1047<br>J1127<br>J1129<br>J1188 |
|                        | Saratoga 138/69 kV transformer                 | ATC            | P23                                                                      | J986                                               |
|                        | Rocky Run – Werner West 345 kV line            | ATC            | P12<br>P23                                                               | J986<br>J1085<br>J1101<br>J1121                    |
|                        | North Appleton – Werner West 345 kV line       | ATC            | P12                                                                      | J986<br>J1085<br>J1101<br>J1121                    |
|                        | Lublin – Lublin Pump 69 kV line                | DPC            | P12<br>P23                                                               | J1085<br>J1121                                     |
|                        | Lublin Tap – Lublin Pump 69 kV line            | DPC            | P12<br>P23                                                               | J1085<br>J1121                                     |
|                        | Stone Lake 345/161 kV transformer              | XEL            | P12<br>P23                                                               | J1085<br>J1121                                     |

<sup>1</sup> The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

### 2.3.5 Network Upgrades Identified in ERIS Analysis

Based on the steady-state analyses, the worst loading of each facility under "No Load Loss Allowed" NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3.3-1. Potential Network Upgrades are also included. According to ATC's Transmission Planning Assessment Practices as listed in Appendix B, MISO generator interconnection studies shall utilize a five percent Transmission Reliability Margin (TRM) reduction in normal and emergency ratings for all facilities inside the ATC system. Good faith Cost Estimates of the ERIS thermal Network Upgrades identified in the steady-state analysis for the 2023 scenarios are listed in Table 2.3.3-2. Detailed cost allocations are provided in Section 9.

Table 2.3.5-1 – ERS Network Upgrades Identified to Address  
MISO Steady-State Injection Constraints

| Steady-State Injection Constraint              | Responsible Generator(s)                           | Facility Owner | Study Case                   | Applicable Rating (MVA) <sup>1</sup> | Worst Loading (%) | "No Load Loss Allowed" NERC TPL Planning Event | Network Upgrades                                                                                   |
|------------------------------------------------|----------------------------------------------------|----------------|------------------------------|--------------------------------------|-------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Burlington – North Lake Geneva Tap 138 kV line | J1042<br>J1044<br>J1047<br>J1127<br>J1129<br>J1188 | ATC            | 23SUM, Batteries Charging    | 225                                  | 136.7             |                                                | Burlington – North Lake Geneva Tap 138 kV line, reconductor                                        |
| Sand Lake 138/69 kV transformer                | J986                                               | ATC            | 23SUM, Batteries Discharging | 60                                   | 109.6             |                                                | Sand Lake 138/69 kV transformer, replacement                                                       |
| Wautoma 138/69 kV transformer                  | J986<br>J1002                                      | ATC            | 23SUM, Batteries Charging    | 65                                   | 133.1             |                                                | Wautoma 138/59 kV transformer, replacement                                                         |
| J986 POI – Port Edwards 138 kV line            | J986<br>J1002                                      | ATC            | 23SUM, Batteries Discharging | 153                                  | 107.0             |                                                | J986 POI – Port Edwards 138 kV line, uprate                                                        |
| Concord – Rubicon 138 kV line                  | J1051<br>J1053                                     | ATC            | 23SUM, Batteries Discharging | 335                                  | 102.3             |                                                | Concord – Bark River 138 kV, new line;<br>Concord – Crawfish River – Jefferson 138 kV line, uprate |
| Summit 2-3 138 kV bus tie                      | J1051<br>J1053<br>J1154                            | ATC            | 23SUM, Batteries Discharging | 381                                  | 96.6              |                                                |                                                                                                    |
| Cooney – Summit 138 kV line                    | J1003<br>J1051<br>J1053<br>J1154                   | ATC            | 23SUM, Batteries Discharging | 335                                  | 114.4             |                                                |                                                                                                    |
| Concord – Cooney 138 kV line                   | J1003<br>J1051<br>J1053<br>J1154                   | ATC            | 23SUM, Batteries Discharging | 402                                  | 103.9             |                                                |                                                                                                    |
| Saratoga 138/69 kV transformer                 | J986                                               | ATC            | 23SH, Batteries Charging     | 239                                  | 100.4             |                                                | Saratoga 138/115 kV transformer, re-tap CT                                                         |
| Rocky Run – Werner West 345 kV line            | J986<br>J1085<br>J1101<br>J1121                    | ATC            | 23SH, Batteries Charging     | 861                                  | 109.5             |                                                | Rocky Run – Werner West 345 kV line, uprate                                                        |
| North Appleton – Werner West 345 kV line       | J986<br>J1085<br>J1101<br>J1121                    | ATC            | 23SH, Batteries Charging     | 826                                  | 97.5              |                                                | North Appleton – Werner West 345 kV line, uprate                                                   |
| Lublin – Lublin Pump 69 kV line                | J1085<br>J1121                                     | DPC            | 23SH, Batteries Charging     | 25                                   | 151.7             |                                                | Lublin – Lublin Pump 69 kV line, uprate                                                            |
| Lublin Tap – Lublin Pump 69 kV line            | J1085<br>J1121                                     | DPC            | 23SH, Batteries Charging     | 25                                   | 145.9             |                                                | Lublin Tap – Lublin Pump 69 kV line, uprate                                                        |
| Stone Lake 345/161 kV transformer              | J1085<br>J1121                                     | XCEL           | 23SH, Batteries Charging     | 515.2                                | 113.4             |                                                | Stone Lake 345/161 kV transformer, replacement                                                     |

<sup>1</sup> The rating of the Injection Constraint without TRM.

**Table 2.3.5-2 – ERIS Network Upgrades and Cost Estimates**

| Steady-State Injection Constraint              | Facility Owner | Network Upgrade                                                                                    | Cost (\$) <sup>1,2,3</sup> |
|------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------|----------------------------|
| Burlington – North Lake Geneva Tap 138 kV line | ATC            | Burlington – North Lake Geneva Tap 138 kV line, reconductor <sup>4</sup>                           | 10,692,501                 |
| Sand Lake 138/69 kV transformer                | ATC            | Sand Lake 138/69 kV transformer, replacement                                                       | 3,105,696                  |
| Wautoma 138/69 kV transformer                  | ATC            | Wautoma 138/59 kV transformer, replacement                                                         | 2,859,187                  |
| J986 POI – Port Edwards 138 kV line            | ATC            | J986 POI – Port Edwards 138 kV line, uprate                                                        | 584,677                    |
| Concord – Rubicon 138 kV line                  | ATC            | Concord – Bark River 138 kV, new line;<br>Concord – Crawfish River – Jefferson 138 kV line, uprate | 33,818,778                 |
| Summit 2-3 138 kV bus tie                      | ATC            |                                                                                                    |                            |
| Cooney – Summit 138 kV line                    | ATC            |                                                                                                    |                            |
| Concord – Cooney 138 kV line                   | ATC            |                                                                                                    |                            |
| Saratoga 138/69 kV transformer                 | ATC            | Saratoga 138/69 kV transformer, re-tap CT                                                          | 114,149                    |
| Rocky Run – Werner West 345 kV line            | ATC            | Rocky Run – Werner West 345 kV line, uprate                                                        | 330,135                    |
| North Appleton – Werner West 345 kV line       | ATC            | North Appleton – Werner West 345 kV line, uprate                                                   | 597,741                    |
| Lublin – Lublin Pump 69 kV line                | DPC            | Lublin – Lublin Pump 69 kV line, uprate                                                            | 364,865                    |
| Lublin Tap – Lublin Pump 69 kV line            | DPC            | Lublin Tap – Lublin Pump 69 kV line, uprate                                                        | 4,135,135                  |
| Stone Lake 345/161 kV transformer              | XCEL           | Stone Lake 345/161 kV transformer, replacement <sup>5</sup>                                        | 3,757,200                  |

<sup>1</sup> All Network Upgrades were estimated on the generator ISD dollars.

<sup>2</sup> ATC Network Upgrades included a 10% contingency.

<sup>3</sup> No contingency was included for the Stone Lake transformer upgrade project according to Xcel Energy.

<sup>4</sup> Cost Estimate is the APR-18 ERIS cost (10,972,884) minus the AUG-17 NRIS cost (\$280,383) to uprate the line

<sup>5</sup> Cost Estimate is the APR-18 ERIS cost (\$7,655,700) minus the AUG-17 NRIS cost (\$3,898,500) to replace the transformer

### 2.3.6 Network Upgrade Alternatives Considered

All but one of the ERIS network upgrades identified in Table 2.3.3-2 are direct upgrades of the ERIS thermal constraint facilities to ATC, DPC, and XCEL design standards and considered as least-cost solutions. Therefore, no other alternatives were examined at for those solutions.

For the Concord – Bark River 138 kV new line, ATC will consider other alternatives in Phase 2.

### 2.3.7 Potential Operating Restriction

Operating restriction analysis will be performed as part of MISO DPP April 2018 Phase 2.

### 2.3.8 Additional Studies for J1085 and J1121

Due to unique concerns with the location of the J1085 and J1121 POI, ATC performed the following additional studies. The six study reports were included in Appendix K.

### 1. Shunt Reactor Need and Sizing Study

This study reviewed the need for a line-connected shunt reactor on the Stone Lake – J1085/J1121 POI 345 kV line at the J1085/J1121 POI and determined the required reactive power capability based on steady state voltage and transient studies of coupled line resonance and delayed current zeros [REDACTED]. Study results show that an 80 Mvar shunt reactor is needed to safely energize the line under a variety of system conditions and that any breaker that could be used to energize the line must be equipped with pre-insertion resistors.

**Table 2.3.8-1 – ERIIS Network Upgrades Identified**

| System Constraints                                      | Facility Owner | Network Upgrade                                                         | ISD (For Cost Estimate Only) | Cost (\$) <sup>1,2</sup> |
|---------------------------------------------------------|----------------|-------------------------------------------------------------------------|------------------------------|--------------------------|
| Overvoltage on J1085/J1121 POI – Stone Lake 345 kV line | ATC            | J1085/J1121 – Stone Lake 345 kV, install 80 Mvar line-connected reactor | 09/01/2021                   | 4,356,036                |

<sup>1</sup> All Network Upgrades were estimated on the earliest ISD dollars of responsible generator.

<sup>2</sup> ATC Network Upgrades included a 10% contingency.

### 2. Steady State Voltage Stability Study of MWEX

This analysis was performed on the 2023 shoulder charging study case to determine if the April 2018 DPP cycle, including J1085 and J1121, could cause or contribute to voltage stability violations on MWEX (Minnesota Wisconsin Export Interface).

The study results show that the Post-DPP scenario is voltage stable and meets ATC voltage stability criteria. Therefore, voltage stability related Network Upgrades will not be assigned to any Interconnection Customers.

### 3. Steady State Voltage Stability Sensitivity Study of J1085/J1121 with Local P6 Outages

This analysis was performed on the 2023 shoulder charging study case to determine if J1085 and J1121 could cause voltage stability violations under P6 (N-1-1) outage conditions.

The results show that a voltage collapse could occur with J732, J1085, and J1121 operating at less than their maximum requested capabilities under select NERC Category P6 contingencies. ATC Planning Criteria does not require Network Upgrades to eliminate violations of planning criteria with NERC Category P6 contingencies, however operating restrictions may be required under certain prior outage scenarios as shown in the table below.

**Table 2.3.8-2: Operating Restriction Summary**

| Prior Outage | Potential J1085/J1121<br>Combined Output Restriction<br>(MW) |
|--------------|--------------------------------------------------------------|
|              | 210                                                          |
|              | 210                                                          |
|              | 342                                                          |

#### 4. Tripping Systems Need and Redesign at the [REDACTED] [REDACTED] Substations

The existing Cross Tripping System (CTS) and Over-Voltage Tripping System (OVTS) requirements at [REDACTED].

With the proposed interconnection of [REDACTED], these tripping systems were re-evaluated for possible modifications.

[REDACTED]

### 3.0 STABILITY ANALYSIS

Stability analysis will be performed as part of MISO DPP April 2018 Phase 2.

### 4.0 SHORT CIRCUIT ANALYSIS

Short circuit analysis will be performed as part of MISO DPP April 2018 Phase 2.

### 5.0 TRANSFORMER ENERGIZATION ANALYSIS

#### 5.1 Transformer Energization Study Results

Transformer Energization Analysis were performed based on ATC transformer initial energization criteria as described in Appendix A. The 2023 shoulder model (batteries charging) was used with all identified ERIS and NRIS Network Upgrades included. The results are summarized in Table 5.1-1. No constraints were found.

**Table 5.1-1 – Inrush Calculations Using Shoulder Model**

| Generation Project | Transformer(s) |          | PSSE POI | Fault       | Vmin Inrush (pu) |            |
|--------------------|----------------|----------|----------|-------------|------------------|------------|
|                    | Number         | Windings | Bus #    | Current (A) | Raw              | Multiplier |
| J986               | 1              | 2        | 89864    | 6392        | 0.7307           | 0.8404     |
| J1000              | 1              | 2        | 699020   | 8976.4      | 0.9401           | 1.0811     |
| J1002              | 1              | 2        | 699235   | 5412.7      | 0.8277           | 0.9519     |
| J1003              | 1              | 2        | 698222   | 7133.3      | 0.8474           | 0.9745     |
| J1009              | 2              | 3        | 40095    | 22526.9     | 0.9549           | 1.0982     |
| J1010              | 2              | 3        | 999901   | 14078.7     | 0.9298           | 1.0692     |
| J1011              | 2              | 3        | 999901   | 14078.7     | 0.9310           | 1.0706     |
| J1042              | 1              | 3        | 699085   | 11542.3     | 0.8164           | 0.9389     |
| J1044              | 1              | 3        | 699360   | 11968.6     | 0.8269           | 0.9509     |
| J1047              | 1              | 3        | 699360   | 11968.6     | 0.8193           | 0.9422     |
| J1051              | 1              | 2        | 40514    | 12189.1     | 0.8685           | 0.9988     |
| J1053              | 2              | 2        | 40514    | 12189.1     | 0.8685           | 0.9988     |
| J1085              | 2              | 3        | 40854    | 8808.3      | 0.8923           | 1.0261     |
| J1101              | 1              | 2        | 699620   | 19785.6     | 0.9847           | 1.1323     |
| J1104              | 1              | 2        | 41044    | 8310.6      | 0.8520           | 0.9798     |
| J1121              | 1              | 3        | 40854    | 8912.5      | 0.8934           | 1.0274     |
| J1127              | 1              | 2        | 41274    | 6177.8      | 0.9109           | 1.0475     |
| J1129              | 1              | 2        | 41294    | 21826.5     | 0.9647           | 1.1094     |
| J1153              | 1              | 2        | 699288   | 6355.9      | 0.7923           | 0.9111     |
| J1154              | 1              | 2        | 699340   | 12208.1     | 0.8252           | 0.9490     |
| J1171              | 1              | 2        | 699270   | 8634.5      | 0.7574           | 0.8711     |
| J1183              | 1              | 2        | 693701   | 3207.1      | 0.8177           | 0.9404     |
| J1188              | 1              | 2        | 699104   | 7220.6      | 0.8587           | 0.9875     |

## 6.0 LOW SHORT CIRCUIT STRENGTH SCREENING ANALYSIS

ATC performed a low short circuit strength screening analysis for all inverter-based resources in the DPP-2018-April East (ATC) Area Phase 1. The methodology for the low short circuit strength screening analysis is detailed in Appendix B. PSCAD analysis will be performed as part of MISO DPP April 2018 Phase 2.

### 6.1 Low Short Circuit Strength Screening Results

The low short circuit strength screening analysis included classic Short Circuit Ratio (SCR) analysis with results shown in Tables 6.1-1, 6.1-2, and 6.1-3. Classic SCRs at POIs that are lower than the designed minimum SCRs (if provided) of the inverters are highlighted in red.

**Table 6.1-1: SCR Analysis Results for System Intact**

| MISO Queue #         | POI Bus # | Capacity (MW) | Short Circuit Capacity (MVA) | Short Circuit Ratio at POI | Designed Minimum SCR |
|----------------------|-----------|---------------|------------------------------|----------------------------|----------------------|
| J986                 | 89864     | 150           | 1559.78                      | 10.40                      | 2                    |
| J1000                | 699020    | 50            | 2168.95                      | 43.38                      | 3                    |
| J1002                | 699235    | 99            | 1332.41                      | 13.46                      | 3                    |
| J1003                | 698222    | 50            | 877.36                       | 17.55                      | 3                    |
| J1009                | 693481    | 400           | 13477.26                     | 33.69                      | 2.5                  |
| J1010/J1011          | 999901    | 800           | 8397.20                      | 10.50                      | 2.5                  |
| J1042                | 699085    | 200           | 2701.34                      | 13.51                      | 2.5                  |
| J1044/J1047          | 699360    | 400           | 2720.71                      | 6.80                       | 2.5                  |
| J1051/J1053          | 698887    | 350           | 2939.27                      | 8.40                       | 2                    |
| J886/J1101           | 699620    | 170           | 4739.39                      | 27.88                      | 2.5                  |
| J1104                | 699319    | 100           | 2003.63                      | 20.04                      | 2.5                  |
| J1085/J1121          | 40854     | 500           | 5547.71                      | 11.10                      | 1.7/2.5              |
| J1127                | 41274     | 49.9          | 1528.93                      | 30.58                      | 3                    |
| J1129                | 41294     | 65            | 5221.81                      | 80.34                      | 3                    |
| J1153                | 699288    | 150           | 1528.27                      | 10.19                      | 2                    |
| J1154/J818           | 698884    | 224           | 2942.93                      | 13.14                      | 2                    |
| J1171                | 699270    | 100           | 2080.23                      | 20.80                      | 3                    |
| J1183/Garden Corners | 693701    | 110           | 806.63                       | 7.33                       | 4.24                 |
| J1188                | 699104    | 50            | 888.17                       | 17.76                      | 3                    |

**Table 6.1-2: SCR Analysis Results for Worst N-1 Contingency**

| MISO Queue #         | POI Bus # | Capacity (MW) | Short Circuit Capacity (MVA) | Short Circuit Ratio at POI | Designed Minimum SCR |
|----------------------|-----------|---------------|------------------------------|----------------------------|----------------------|
| J986                 | 89864     | 150           | 451.93                       | 3.01                       | 2                    |
| J1000                | 699020    | 50            | 676.37                       | 13.53                      | 3                    |
| J1002                | 699235    | 99            | 803.29                       | 8.11                       | 3                    |
| J1003                | 698222    | 50            | 506.26                       | 10.13                      | 3                    |
| J1009                | 693481    | 400           | 9453.29                      | 23.63                      | 2.5                  |
| J1010/J1011          | 999901    | 800           | 4564.39                      | 5.71                       | 2.5                  |
| J1042                | 699085    | 200           | 1003.52                      | 5.02                       | 2.5                  |
| J1044/J1047          | 699360    | 400           | 1674.80                      | 4.19                       | 2.5                  |
| J1051/J1053          | 698887    | 350           | 1518.43                      | 4.34                       | 2                    |
| J886/J1101           | 699620    | 170           | 3143.93                      | 18.49                      | 2.5                  |
| J1104                | 699319    | 100           | 1018.86                      | 10.19                      | 2.5                  |
| J1085/J1121          | 40854     | 500           | 1809.21                      | 3.62                       | 1.7/2.5              |
| J1127                | 41274     | 49.9          | 486.35                       | 9.73                       | 3                    |
| J1129                | 41294     | 65            | 2573.07                      | 39.59                      | 3                    |
| J1153                | 699288    | 150           | 358.30                       | 2.39                       | 2                    |
| J1154/J818           | 698884    | 224           | 1551.18                      | 6.92                       | 2                    |
| J1171                | 699270    | 100           | 679.40                       | 6.79                       | 3                    |
| J1183/Garden Corners | 693701    | 110           | 507.70                       | 4.62                       | 4.24                 |
| J1188                | 699104    | 50            | 507.92                       | 10.16                      | 3                    |

**Table 6.1-3: SCR Analysis Results for Worst N-1-1 Contingency**

| MISO Queue # | POI Bus # | Capacity (MW) | Short Circuit Capacity (MVA) | Short Circuit Ratio at POI | Designed Minimum SCR |
|--------------|-----------|---------------|------------------------------|----------------------------|----------------------|
| J986         | 89864     | 150           | 160.39                       | 1.07                       | 2                    |
| J1000        | 699020    | 50            | 520.97                       | 10.42                      | 3                    |
| J1002        | 699235    | 99            | 341.47                       | 3.45                       | 3                    |
| J1003        | 698222    | 50            | 156.71                       | 3.13                       | 3                    |
| J1009        | 693481    | 400           | 5588.20                      | 13.97                      | 2.5                  |
| J1010/J1011  | 999901    | 800           | 1740.65                      | 2.18                       | 2.5                  |
| J1042        | 699085    | 200           | 892.20                       | 4.46                       | 2.5                  |
| J1044/J1047  | 699360    | 400           | 800.40                       | 2.00                       | 2.5                  |
| J1051/J1053  | 698887    | 350           | 1443.67                      | 4.12                       | 2                    |
| J886/J1101   | 699620    | 170           | 1488.84                      | 8.76                       | 2.5                  |
| J1104        | 699319    | 100           | 772.85                       | 7.73                       | 2.5                  |
| J1085/J1121  | 40854     | 500           | 1368.00                      | 2.74                       | 1.7/2.5              |

|                         |        |      |         |       |      |
|-------------------------|--------|------|---------|-------|------|
| J1127                   | 41274  | 49.9 | 232.90  | 4.66  | 3    |
| J1129                   | 41294  | 65   | 1253.21 | 19.28 | 3    |
| J1153                   | 699288 | 150  | 355.19  | 2.37  | 2    |
| J1154/J818              | 698884 | 224  | 837.31  | 3.74  | 2    |
| J1171                   | 699270 | 100  | 490.44  | 4.90  | 3    |
| J1183/Garden<br>Corners | 693701 | 110  | 486.85  | 4.43  | 4.24 |
| J1188                   | 699104 | 50   | 199.43  | 3.99  | 3    |

It was determined that several groups of inverter-based resources were placed sufficiently close together to require Weighted Short Circuit Ratio (WSCR) analysis. The results for these WSCR analyses are shown in Tables 6.1-4, 6.1-5, and 6.1-6.

**Table 6.1-4: WSCR Analysis Results for System Intact**

| Generator Group                                       | Weighted SCR |
|-------------------------------------------------------|--------------|
| J986, J1002                                           | 5.90         |
| Quiltblock, J807, J819, J855, J947, J1000             | 2.93         |
| J855, J870, J871, J947, J1000                         | 3.14         |
| Quiltblock, J807, J819, J855, J870, J871, J947, J1000 | 2.10         |
| J878, J1010, J1011                                    | 7.77         |
| J878, J1010, J1011, J1042, J1044, J1047               | 3.67         |
| J1042, J1044, J1047                                   | 4.52         |
| Quiltblock, J807, J819, J947, J1127                   | 5.48         |
| Forward Energy Center, J1171                          | 9.91         |

**Table 6.1-5: WSCR Analysis Results for Worst N-1 Contingency**

| Generator Group                                       | Weighted SCR |
|-------------------------------------------------------|--------------|
| J986, J1002                                           | 2.59         |
| Quiltblock, J807, J819, J855, J947, J1000             | 2.11         |
| J855, J870, J871, J947, J1000                         | 2.79         |
| Quiltblock, J807, J819, J855, J870, J871, J947, J1000 | 1.44         |
| J878, J1010, J1011                                    | 4.55         |
| J878, J1010, J1011, J1042, J1044, J1047               | 2.40         |
| J1042, J1044, J1047                                   | 3.34         |
| Quiltblock, J807, J819, J947, J1127                   | 4.40         |
| Forward Energy Center, J1171                          | 3.41         |

**Table 6.1-6: WSCR Analysis Results for Worst N-1-1 Contingency**

| Generator Group                                       | Weighted SCR |
|-------------------------------------------------------|--------------|
| J986, J1002                                           | 1.35         |
| Quiltblock, J807, J819, J855, J947, J1000             | 1.50         |
| J855, J870, J871, J947, J1000                         | 2.13         |
| Quiltblock, J807, J819, J855, J870, J871, J947, J1000 | 1.13         |
| J878, J1010, J1011                                    | 1.98         |
| J878, J1010, J1011, J1042, J1044, J1047               | 1.38         |
| J1042, J1044, J1047                                   | 2.23         |
| Quiltblock, J807, J819, J947, J1127                   | 2.26         |
| Forward Energy Center, J1171                          | 2.66         |

Application of the ATC Low Short Circuit Strength Guideline determined that J986, J1000, J1002, J1010, J1011, J1042, J1044, J1047, J1085, J1121, J1127, J1153, and J1183 will all require detailed PSCAD studies to be performed to assess stability. Table 6.1-7 below summarizes the rationale for the PSCAD analysis decision for each queue request.

**Table 6.1-7: Required PSCAD Studies and Rationale**

| MISO Queue #         | Low SCR | Low WSCR | East of Arnold Substation | Operational / Previous Study Experience    | PSCAD Study Required? |
|----------------------|---------|----------|---------------------------|--------------------------------------------|-----------------------|
| J986                 | Yes     | Yes      | No                        | No                                         | Yes                   |
| J1000                | No      | Yes      | No                        | No                                         | Yes                   |
| J1002                | No      | Yes      | No                        | No                                         | Yes                   |
| J1003                | No      | No       | No                        | No                                         | No                    |
| J1009                | No      | No       | No                        | No                                         | No                    |
| J1010/J1011          | Yes     | Yes      | No                        | No                                         | Yes                   |
| J1042                | No      | Yes      | No                        | No                                         | Yes                   |
| J1044/J1047          | Yes     | Yes      | No                        | No                                         | Yes                   |
| J1051/J1053          | No      | No       | No                        | No                                         | No                    |
| J886/J1101           | No      | No       | No                        | No                                         | No                    |
| J1104                | No      | No       | No                        | No                                         | No                    |
| J1085/J1121          | No      | No       | No                        | Known Voltage Stability Issues in the Area | Yes                   |
| J1127                | No      | No       | No                        | Known Weak Area from DPP-AUG-17            | Yes                   |
| J1129                | No      | No       | No                        | No                                         | No                    |
| J1153                | Yes     | No       | No                        | No                                         | Yes                   |
| J1154/J818           | No      | No       | No                        | No                                         | No                    |
| J1171                | No      | No       | No                        | No                                         | No                    |
| J1183/Garden Corners | No      | No       | Yes                       | No                                         | Yes*                  |

|       |    |    |    |    |    |
|-------|----|----|----|----|----|
| J1188 | No | No | No | No | No |
|-------|----|----|----|----|----|

\*A PSCAD study of J1183/Garden Corners is required per ATC Low Short Circuit Strength Guidelines due to the proximity of the plant to the Mackinac HVDC device. However, ATC is still waiting for an updated PSCAD model of the Mackinac HVDC device from the manufacturer so the study may not be able to start in Phase 2 of DPP-APR-18.

## 7.0 AFFECTED SYSTEM ANALYSIS

Affected system analyses will be performed as part of MISO DPP April 2018 Phase 2.

## 8.0 DELIVERABILITY STUDY

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at

[https://cdn.misoenergy.org/Generator\\_Deliverability\\_Study\\_Methodology108139.pdf](https://cdn.misoenergy.org/Generator_Deliverability_Study_Methodology108139.pdf)

### 8.1 Study Summary

The summary of MISO deliverability results based on the 2023 summer peak study model is shown in the following tables.

Table 8.1-1 below lists the deliverability results with ERIS upgrades included in the NRIS analysis. Minimum NR Deliverable is the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew. Maximum NR Deliverable is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction.

**Table 8.1-1 – NRIS Analysis Summary**

| <b>MISO Queue #</b> | <b>Area</b> | <b>NR Tested</b> | <b>Minimum NR Deliverable (MW)</b> | <b>Maximum NR Deliverable (MW)</b> |
|---------------------|-------------|------------------|------------------------------------|------------------------------------|
| J986                | ATC         | 149.76           | 149.76                             | 149.76                             |
| J1000               | ATC         | 50               | 50                                 | 50                                 |
| J1002               | ATC         | 99               | 99                                 | 99                                 |
| J1003               | ATC         | 50               | 50                                 | 50                                 |
| J1009               | ATC         | 400              | 400                                | 400                                |
| J1010               | ATC         | 400              | 389.56                             | 396.28                             |
| J1011               | ATC         | 400              | 389.56                             | 396.28                             |
| J1042               | ATC         | 200              | 0                                  | 80.32                              |
| J1044               | ATC         | 200              | 0                                  | 80.32                              |
| J1047               | ATC         | 200              | 0                                  | 80.32                              |
| J1051               | ATC         | 50               | 50                                 | 50                                 |
| J1053               | ATC         | 300              | 300                                | 300                                |
| J1085               | ATC         | 300              | 300                                | 300                                |
| J1101               | ATC         | 20               | 20                                 | 20                                 |
| J1104               | ATC         | 100              | 13.36                              | 13.36                              |
| J1121               | ATC         | 200              | 200                                | 200                                |
| J1127               | ATC         | 49.9             | 0                                  | 0                                  |
| J1129               | ATC         | 65               | 0                                  | 0                                  |
| J1153               | ATC         | 150              | 150                                | 150                                |
| J1154               | ATC         | 75               | 75                                 | 75                                 |
| J1171               | ATC         | 100              | 100                                | 100                                |
| J1183               | ATC         | 0                | 0                                  | 0                                  |
| J1188               | ATC         | 50               | 0                                  | 0                                  |

Table 8.1-2 below lists all of the NRIS constraints from the deliverability study and the identified NRIS Network Upgrades. Both ERIS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERIS Network Upgrade, only the cost difference between the NRIS upgrade and the ERIS upgrade will be eligible for NRIS Network Upgrade cost allocation. Detailed NRIS Network Upgrade cost allocation calculations are provided in Section 9.

Table 8.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

| NRIS Thermal Constraints                                                                                                             | Required NRIS Network Upgrades                                                                                                                                                                         | Mitigation Type | Responsible Generators                                        | NRIS Network Upgrade Cost Estimates (\$) [a] | ERIS Network Upgrade (Required for Same Constraint) Cost Estimate (\$) [b] | Previous Cycle ERIS/NRIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [c] | Cost Used for NRIS Network Upgrade Allocation (\$) [c] = [a] - [b] |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 41044 J1104 POI 138 699260 BARK RVR 138 1 C                                                                                          | J1104 - Bark River 138 kV line, Reconductor                                                                                                                                                            | NR              | J1104                                                         | 3,479,462                                    | 0                                                                          | 0                                                                                                | 3,479,462                                                          |
| 699319 GERMNTN 138 699380 MAPLE 138 1                                                                                                | Germantown - Maple 138 kV line, Reconductor                                                                                                                                                            | NR              | J1104                                                         | 2,733,155                                    | 0                                                                          | 0                                                                                                | 2,733,155                                                          |
| 694106 ROR BUS1 138 699062 MRE 138 138 1                                                                                             | Rock River - Marine 138 kV line, Reconductor                                                                                                                                                           | NR              | J1127, J1188                                                  | 2,499,649                                    | 0                                                                          | 0                                                                                                | 2,499,649                                                          |
| 698007 DAR 69 69.0 698018 ROB 69 69.0 1                                                                                              | Darlington - Rock Branch 69 kV line, Uprate                                                                                                                                                            | NR              | J1127                                                         | 192,841                                      | 0                                                                          | 0                                                                                                | 192,841                                                            |
| 698028 NOM 69 69.0 698036 MAP 69 69.0 1                                                                                              | North Monroe - Monticello Tap 69 kV line, Uprate                                                                                                                                                       | NR              | J1127                                                         | 41,915                                       | 0                                                                          | 0                                                                                                | 41,915                                                             |
| 698090 BOL 138 138 699086 ELK 138 138 1                                                                                              | Bristol - Elkhorn 138 kV line, Uprate                                                                                                                                                                  | NR              | J1127, J1129, J1188                                           | 193,170                                      | 0                                                                          | 0                                                                                                | 193,170                                                            |
| 699085 NLG 138 138 699360 NLK GV T 138 1                                                                                             | North Lake Geneva - North Lake Geneva Tap 138 kV line, Uprate                                                                                                                                          | NR              | J1042, J1044, J1047, J1127, J1129, J1188                      | 71,805                                       | 0                                                                          | 0                                                                                                | 71,805                                                             |
| 699086 ELK 138 138 699085 NLG 138 138 1                                                                                              | Elkhorn - North Lake Geneva 138 kV line, Uprate                                                                                                                                                        | NR              | J1127, J1129, J1188                                           | 159,914                                      | 0                                                                          | 45,400                                                                                           | 114,514                                                            |
| 699512 UNVRSTY 138 699357 MUKWONGO 138 1                                                                                             | University - Mukwonago 138 kV line, Rebuild                                                                                                                                                            | NR              | J1042, J1044, J1047, J1127, J1129, J1188                      | 24,244,505                                   | 0                                                                          | 0                                                                                                | 24,244,505                                                         |
| 699360 NLK GV T 138 699267 BRLGTN1 138 1<br>699409 PARIS B5 138 693647 BERRYVILLE 138 1<br>699410 PARIS WE 138 699409 PARIS B5 138 Z | North Lake Geneva Tap - Burlington 138 kV line, Rebuild to Double Circuit<br>Sugar Creek - North Lake Geneva Tap 138 kV line, Uprate<br>Whitewater - Bluff Creek Tap - Sugar Creek 138 kV line, Uprate | NR              | J1010, J1011, J1042, J1044, J1047, J1104, J1127, J1129, J1188 | 17,777,542<br>356,565<br>350,275             | 10,692,501<br>0<br>0                                                       | 280,383<br>0<br>0                                                                                | 6,804,658<br>356,565<br>350,275                                    |

8.2 Per Project Summary

8.2.1 J986

|                                                                                                               |                        |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|---------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J986 Deliverable (NRIS) Amount in 2023 Case: 149.76 MW<br>(Conditional on ERIS upgrades and case assumptions) | 149.76 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                               |                        |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)          | Single<br>Deduction    | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                               |                        | 149.76                                                                                                                                                                                                    |                     | 149.76                                                                                                                                                                                                    |           |                                                |                                   |                               |       |

8.2.2 J1000

|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1000 Deliverable (NRIS) Amount in 2023 Case: 50 MW<br>(Conditional on ERIS upgrades and case assumptions) | 50 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                            |                     | 50                                                                                                                                                                                                        |                     | 50                                                                                                                                                                                                        |           |                                                |                                   |                               |       |

8.2.3 J1002

|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1002 Deliverable (NRIS) Amount in 2023 Case: 99 MW<br>(Conditional on ERIS upgrades and case assumptions) | 99 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                            |                     | 99                                                                                                                                                                                                        |                     | 99                                                                                                                                                                                                        |           |                                                |                                   |                               |       |

8.2.4 J1003

|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1003 Deliverable (NRIS) Amount in 2023 Case: 50 MW<br>(Conditional on ERIS upgrades and case assumptions) | 50 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                            |                     | 50                                                                                                                                                                                                        |                     | 50                                                                                                                                                                                                        |           |                                                |                                   |                               |       |

8.2.5 J1009

|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1009 Deliverable (NRIS) Amount in 2023 Case: 400 MW<br>(Conditional on ERIS upgrades and case assumptions) | 400 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                             |                     | 400                                                                                                                                                                                                       |                     | 400                                                                                                                                                                                                       |           |                                                |                                   |                               |       |

8.2.6 J1010

|                                                                                                                |                     |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|----------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1010 Deliverable (NRIS) Amount in 2023 Case: 389.56 MW<br>(Conditional on ERIS upgrades and case assumptions) | 400 MW<br>Requested |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                                |                     |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)           | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated With<br>NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699409 PARIS B5    138   693647 BERRYVILLE    138   1                                                          | 10.44               | 389.56                                                                                                                                                                                                 | 3.72                | 396.28                                                                                                                                                                                                    | 0.1754    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$827,016                         | \$7,511,498                   |       |
|                                                                                                                |                     | 400                                                                                                                                                                                                    |                     | 400                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.7 J1011

|                                                                                                               |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|---------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1011 Deliverable (NRIS) Amount in 2023 Case: 389.56 MW<br>(Conditional on ERS upgrades and case assumptions) | 400 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                               |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)          | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699409 PARIS B5 138 693647 BERRYVILLE 138 1                                                                   | 10.44               | 389.56                                                                                                                                                                                                    | 3.72                | 396.28                                                                                                                                                                                                    | 0.1754    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$827,016                         | \$7,511,498                   |       |
|                                                                                                               |                     | 400                                                                                                                                                                                                       |                     | 400                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.8 J1042

|                                                                                                          |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|----------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1042 Deliverable (NRIS) Amount in 2023 Case: 0 MW<br>(Conditional on ERS upgrades and case assumptions) | 200                 |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                          |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)     | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated With<br>NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY 138 699357 MUKWONGO 138 1                                                                 | 472.33              | 0                                                                                                                                                                                                         | 119.68              | 80.32                                                                                                                                                                                                     | 0.0891    | J1042, J1044, J1047,<br>J1104, J1127, J1129,<br>J1188              | \$6,167,802                       | \$24,244,505                  |       |
| 699360 NLK GV T 138 699267 BRLGTN1 138 1                                                                 | 120.01              | 79.99                                                                                                                                                                                                     | 30.71               | 169.29                                                                                                                                                                                                    | 0.2532    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,537,604                       | \$7,511,498                   |       |
| 699410 PARIS WE 138 699409 PARIS B5 138 Z                                                                | 99.06               | 100.94                                                                                                                                                                                                    | 25.29               | 174.71                                                                                                                                                                                                    | 0.2449    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,537,604                       | \$7,511,498                   |       |
| 699409 PARIS B5 138 693647 BERRYVILLE 138 1                                                              | 30.92               | 169.08                                                                                                                                                                                                    | 1.86                | 198.14                                                                                                                                                                                                    | 0.0592    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,537,604                       | \$7,511,498                   |       |
| 699085 NLG 138 138 699360 NLK GV T 138 1                                                                 | 19.17               | 180.83                                                                                                                                                                                                    | 17.22               | 182.78                                                                                                                                                                                                    | 0.5627    | J1042, J1127, J1129,<br>J1188                                      | \$21,305                          | \$71,805                      |       |
|                                                                                                          |                     | 200                                                                                                                                                                                                       |                     | 200                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.9 J1044

|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|-----------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1044 Deliverable (NRIS) Amount in 2023 Case: 0 MW<br>(Conditional on ERIS upgrades and case assumptions) | 200 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)      | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated With<br>NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY 138 699357 MUKWONGO 138 1                                                                  | 389.95              | 0                                                                                                                                                                                                         | 119.68              | 80.32                                                                                                                                                                                                     | 0.1079    | J1042, J1044, J1047,<br>J1104, J1127, J1129,<br>J1188              | \$7,760,666                       | \$24,244,505                  |       |
| 699360 NLK GV T 138 699085 NLG 138 138 1                                                                  | 89.85               | 110.15                                                                                                                                                                                                    | 44.93               | 155.07                                                                                                                                                                                                    | 0.6356    | J1042, J1127, J1129,<br>J1188                                      | \$24,047                          | \$71,805                      |       |
| 699360 NLK GV T 138 699267 BRLGTN1 138 1                                                                  | 89.51               | 110.49                                                                                                                                                                                                    | 30.71               | 169.29                                                                                                                                                                                                    | 0.3395    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
| 699410 PARIS WE 138 699409 PARIS B5 138 Z                                                                 | 73.63               | 126.37                                                                                                                                                                                                    | 25.29               | 174.71                                                                                                                                                                                                    | 0.3295    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
| 699409 PARIS B5 138 693647 BERRYVILLE 138 1                                                               | 18.47               | 181.53                                                                                                                                                                                                    | 1.86                | 198.14                                                                                                                                                                                                    | 0.0991    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
|                                                                                                           |                     | 200                                                                                                                                                                                                       |                     | 200                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.10 J1047

|                                                                                                          |                     |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|----------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1047 Deliverable (NRIS) Amount in 2023 Case: 0 MW<br>(Conditional on ERS upgrades and case assumptions) | 200 MW<br>Requested |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                          |                     |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)     | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be the<br>amount of service to be granted if<br>none of the projects contributing to<br>this constraint, reduced service<br>levels or withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY 138 699357 MUKWONGO 138 1                                                                 | 389.95              | 0                                                                                                                                                                                                      | 119.68              | 80.32                                                                                                                                                                                                     | 0.1079    | J1042, J1044, J1047,<br>J1104, J1127, J1129,<br>J1188              | \$7,760,666                       | \$24,244,505                  |       |
| 699360 NLK GV T 138 699085 NLG 138 138 1                                                                 | 89.85               | 110.15                                                                                                                                                                                                 | 44.93               | 155.07                                                                                                                                                                                                    | 0.6356    | J1042, J1127, J1129,<br>J1188                                      | \$24,047                          | \$71,805                      |       |
| 699360 NLK GV T 138 699267 BRLGTN1 138 1                                                                 | 89.51               | 110.49                                                                                                                                                                                                 | 30.71               | 169.29                                                                                                                                                                                                    | 0.3395    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
| 699410 PARIS WE 138 699409 PARIS B5 138 Z                                                                | 73.63               | 126.37                                                                                                                                                                                                 | 25.29               | 174.71                                                                                                                                                                                                    | 0.3295    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
| 699409 PARIS B5 138 693647 BERRYVILLE 138 1                                                              | 18.47               | 181.53                                                                                                                                                                                                 | 1.86                | 198.14                                                                                                                                                                                                    | 0.0991    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$1,974,773                       | \$7,511,498                   |       |
|                                                                                                          |                     | 200                                                                                                                                                                                                    |                     | 200                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.11 J1051

|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|-----------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1051 Deliverable (NRIS) Amount in 2023 Case: 50 MW<br>(Conditional on ERS upgrades and case assumptions) | 50 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)      | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                           |                     | 50                                                                                                                                                                                                        |                     | 50                                                                                                                                                                                                        |           |                                                |                                   |                               |       |

8.2.12 J1053

|                                                                                                             |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
|-------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------|--------------------------------|----------------------------|-------|
| J1053 Deliverable (NRIS) Amount in 2023 Case: 300 MW<br>(Conditional on ERIS upgrades and case assumptions) | 300 MW Requested |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
|                                                                                                             |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single Deduction | Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew | Shared Deduction | Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction | Max DF | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project | Total NRIS Cost of Upgrade | Notes |
|                                                                                                             |                  | 300                                                                                                                                                                                     |                  | 300                                                                                                                                                                                     |        |                                          |                                |                            |       |

8.2.13 J1085

|                                                                                                             |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
|-------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------|--------------------------------|----------------------------|-------|
| J1085 Deliverable (NRIS) Amount in 2023 Case: 300 MW<br>(Conditional on ERIS upgrades and case assumptions) | 300 MW Requested |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
|                                                                                                             |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single Deduction | Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew | Shared Deduction | Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction | Max DF | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project | Total NRIS Cost of Upgrade | Notes |
|                                                                                                             |                  | 300                                                                                                                                                                                     |                  | 300                                                                                                                                                                                     |        |                                          |                                |                            |       |

8.2.14 J1101

|                                                                                                            |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
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| J1101 Deliverable (NRIS) Amount in 2023 Case: 20 MW<br>(Conditional on ERIS upgrades and case assumptions) | 20 MW Requested  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
|                                                                                                            |                  |                                                                                                                                                                                         |                  |                                                                                                                                                                                         |        |                                          |                                |                            |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single Deduction | Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew | Shared Deduction | Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction | Max DF | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project | Total NRIS Cost of Upgrade | Notes |
|                                                                                                            |                  | 20                                                                                                                                                                                      |                  | 20                                                                                                                                                                                      |        |                                          |                                |                            |       |

8.2.15 J1104

|                                                                                                               |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
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| J1104 Deliverable (NRIS) Amount in 2023 Case: 13.36 MW<br>(Conditional on ERIS upgrades and case assumptions) | 100 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
|                                                                                                               |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                    |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)          | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated With<br>NRIS Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 41044 J1104 POI 138 699260 BARK RVR 138 1 C                                                                   | 86.64               | 13.36                                                                                                                                                                                                     | 86.64               | 13.36                                                                                                                                                                                                     | 0.9986    | J1104                                                              | \$3,479,462                       | \$3,479,462                   |       |
| 699319 GERMNTN 138 699380 MAPLE 138 1                                                                         | 74.36               | 25.64                                                                                                                                                                                                     | 74.36               | 25.64                                                                                                                                                                                                     | 0.9986    | J1104                                                              | \$2,733,155                       | \$2,733,155                   |       |
| 699409 PARIS B5 138 693647 BERRYVILLE 138 1                                                                   | 36.46               | 63.54                                                                                                                                                                                                     | 0.93                | 99.07                                                                                                                                                                                                     | 0.0502    | J1010, J1011, J1042,<br>J1044, J1047, 1104,<br>J1127, J1129, J1188 | \$59,341                          | \$7,511,498                   |       |
|                                                                                                               |                     | 100                                                                                                                                                                                                       |                     | 100                                                                                                                                                                                                       |           |                                                                    |                                   |                               |       |

8.2.16 J1121

|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1121 Deliverable (NRIS) Amount in 2023 Case: 200 MW<br>(Conditional on ERIS upgrades and case assumptions) | 200 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                             |                     | 200                                                                                                                                                                                                       |                     | 200                                                                                                                                                                                                       |           |                                                |                                   |                               |       |

8.2.17 J1127

|                                                                                                           |                      |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|-----------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1127 Deliverable (NRIS) Amount in 2023 Case: 0 MW<br>(Conditional on ERIS upgrades and case assumptions) | 49.9 MW<br>Requested |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|                                                                                                           |                      |                                                                                                                                                                                                        |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)      | Single<br>Deduction  | Minimum Level of Service<br>Attainable (MW) *This would be the<br>amount of service to be granted if<br>none of the projects contributing to<br>this constraint, reduced service<br>levels or withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint                        | NRIS Cost<br>Allocated to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY 138 699357 MUKWONGO 138 1                                                                  | 745.2                | 0                                                                                                                                                                                                      | 25.21               | 24.69                                                                                                                                                                                                     | 0.0752    | J1042, J1044,<br>J1047, J1127,<br>J1129, J1188                        | \$819,464                         | \$24,244,505                  |       |
| 699360 NLK GV T 138 699267 BRLGTN1 138 1                                                                  | 411.02               | 0                                                                                                                                                                                                      | 7.66                | 42.24                                                                                                                                                                                                     | 0.0739    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$99,903                          | \$7,511,498                   |       |
| 699410 PARIS WE 138 699409 PARIS B5 138 Z                                                                 | 338.69               | 0                                                                                                                                                                                                      | 6.31                | 43.59                                                                                                                                                                                                     | 0.0716    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$99,903                          | \$7,511,498                   |       |
| 698090 BOL 138 138 699086 ELK 138 138 1                                                                   | 221.75               | 0                                                                                                                                                                                                      | 75.23               | 0                                                                                                                                                                                                         | 0.0714    | J1127, J1129, J1188                                                   | \$65,562                          | \$193,170                     |       |
| 699086 ELK 138 138 699085 NLG 138 138 1                                                                   | 154.91               | 0                                                                                                                                                                                                      | 50.49               | 0                                                                                                                                                                                                         | 0.1186    | J1127, J1129, J1188                                                   | \$37,355                          | \$114,514                     |       |
| 699085 NLG 138 138 699360 NLK GV T 138 138 1                                                              | 128.78               | 0                                                                                                                                                                                                      | 4.3                 | 45.6                                                                                                                                                                                                      | 0.0838    | J1042, J1127,<br>J1129, J1188                                         | \$790                             | \$71,805                      |       |
| 694106 ROR BUS1 138 699062 MRE 138 138 1                                                                  | 93.09                | 0                                                                                                                                                                                                      | 55.6                | 0                                                                                                                                                                                                         | 0.1336    | J1127, J1188                                                          | \$1,492,541                       | \$2,499,649                   |       |
| 698007 DAR 69 69.0 698018 ROB 69 69.0 1                                                                   | 16.25                | 33.65                                                                                                                                                                                                  | 16.25               | 33.65                                                                                                                                                                                                     | 0.0666    | J1127                                                                 | \$192,841                         | \$192,841                     |       |
| 698028 NOM 69 69.0 698036 MAP 69 69.0 1                                                                   | 7.28                 | 42.62                                                                                                                                                                                                  | 7.28                | 42.62                                                                                                                                                                                                     | 0.1966    | J1127                                                                 | \$41,915                          | \$41,915                      |       |
|                                                                                                           |                      | 49.9                                                                                                                                                                                                   |                     | 49.9                                                                                                                                                                                                      |           |                                                                       |                                   |                               |       |

8.2.18 J1129

|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|-----------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1129 Deliverable (NRIS) Amount in 2023 Case: 0 MW<br>(Conditional on ERIS upgrades and case assumptions) | 65 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|                                                                                                           |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)      | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY    138   699357 MUKWONGO    138   1                                                        | 775.4               | 0                                                                                                                                                                                                         | 32.84               | 32.16                                                                                                                                                                                                     | 0.0722    | J1042, J1044,<br>J1047, J1127,<br>J1129, J1188                        | \$1,027,967                       | \$24,244,505                  |       |
| 699360 NLK GV T    138   699267 BRLGTN1    138   1                                                        | 419.67              | 0                                                                                                                                                                                                         | 9.98                | 55.02                                                                                                                                                                                                     | 0.0724    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$127,695                         | \$7,511,498                   |       |
| 699410 PARIS WE    138   699409 PARIS B5    138   Z                                                       | 346.88              | 0                                                                                                                                                                                                         | 8.22                | 56.78                                                                                                                                                                                                     | 0.0699    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$127,695                         | \$7,511,498                   |       |
| 698090 BOL 138    138   699086 ELK 138    138   1                                                         | 234.71              | 0                                                                                                                                                                                                         | 98                  | 0                                                                                                                                                                                                         | 0.0674    | J1127, J1129, J1188                                                   | \$80,648                          | \$193,170                     |       |
| 699086 ELK 138    138   699085 NLG 138    138   1                                                         | 157.01              | 0                                                                                                                                                                                                         | 65.77               | 0                                                                                                                                                                                                         | 0.117     | J1127, J1129, J1188                                                   | \$47,947                          | \$114,514                     |       |
| 699085 NLG 138    138   699360 NLK GV T    138   1                                                        | 125.13              | 0                                                                                                                                                                                                         | 5.6                 | 59.4                                                                                                                                                                                                      | 0.0862    | J1042, J1127,<br>J1129, J1188                                         | \$1,056                           | \$71,805                      |       |
|                                                                                                           |                     | 65                                                                                                                                                                                                        |                     | 65                                                                                                                                                                                                        |           |                                                                       |                                   |                               |       |

8.2.19 J1153

|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1153 Deliverable (NRIS) Amount in 2023 Case: 150 MW<br>(Conditional on ERIS upgrades and case assumptions) | 150 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                             |                     | 150                                                                                                                                                                                                       |                     | 150                                                                                                                                                                                                       |           |                                                |                                   |                               |       |

8.2.20 J1154

|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1154 Deliverable (NRIS) Amount in 2023 Case: 75 MW<br>(Conditional on ERIS upgrades and case assumptions) | 75 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                            |                     | 75                                                                                                                                                                                                        |                     | 75                                                                                                                                                                                                        |           |                                                |                                   |                               |       |

8.2.21 J1171

|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1171 Deliverable (NRIS) Amount in 2023 Case: 100 MW<br>(Conditional on ERIS upgrades and case assumptions) | 100 MW<br>Requested |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
|                                                                                                             |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)        | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
|                                                                                                             |                     | 100                                                                                                                                                                                                       |                     | 100                                                                                                                                                                                                       |           |                                                |                                   |                               |       |

8.2.22 J1188

|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------|-----------------------------------|-------------------------------|-------|
| J1188 Deliverable (NRIS) Amount in 2023 Case: 50 MW<br>(Conditional on ERIS upgrades and case assumptions) | 50 MW<br>Requested  |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
|                                                                                                            |                     |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                           |           |                                                                       |                                   |                               |       |
| Next Constraint for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS)       | Single<br>Deduction | Minimum Level of Service<br>Attainable (MW) *This would be<br>the amount of service to be<br>granted if none of the projects<br>contributing to this constraint,<br>reduced service levels or<br>withdrew | Shared<br>Deduction | Maximum level of service<br>Attainable (MW) *This would be<br>the service level attainable if all<br>projects contributing to the<br>constraint, reduced their<br>requested NR by the shared<br>deduction | Max<br>DF | Projects Associated<br>With NRIS<br>Constraint                        | NRIS Cost Allocated<br>to Project | Total NRIS Cost<br>of Upgrade | Notes |
| 699512 UNVRSTY    138   699357 MUKWONGO    138   1                                                         | 865.13              | 0                                                                                                                                                                                                         | 25.27               | 24.73                                                                                                                                                                                                     | 0.0648    | J1042, J1044,<br>J1047, J1127,<br>J1129, J1188                        | \$707,940                         | \$24,244,505                  |       |
| 699360 NLK GV T    138   699267 BRLGTN1    138   1                                                         | 496.33              | 0                                                                                                                                                                                                         | 7.68                | 42.32                                                                                                                                                                                                     | 0.0612    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$83,378                          | \$7,511,498                   |       |
| 699410 PARIS WE    138   699409 PARIS B5    138   Z                                                        | 407.81              | 0                                                                                                                                                                                                         | 6.32                | 43.68                                                                                                                                                                                                     | 0.0595    | J1010, J1011,<br>J1042, J1044,<br>J1047, 1104, J1127,<br>J1129, J1188 | \$83,378                          | \$7,511,498                   |       |
| 698090 BOL 138    138   699086 ELK 138    138   1                                                          | 309.99              | 0                                                                                                                                                                                                         | 75.39               | 0                                                                                                                                                                                                         | 0.051     | J1127, J1129, J1188                                                   | \$46,960                          | \$193,170                     |       |
| 699086 ELK 138    138   699085 NLG 138    138   1                                                          | 198.23              | 0                                                                                                                                                                                                         | 50.59               | 0                                                                                                                                                                                                         | 0.0926    | J1127, J1129, J1188                                                   | \$29,213                          | \$114,514                     |       |
| 699085 NLG 138    138   699360 NLK GV T    138   1                                                         | 183.48              | 0                                                                                                                                                                                                         | 4.31                | 45.69                                                                                                                                                                                                     | 0.0588    | J1042, J1127,<br>J1129, J1188                                         | \$560                             | \$71,805                      |       |
| 694106 ROR BUS1    138   699062 MRE 138    138   1                                                         | 138.34              | 0                                                                                                                                                                                                         | 55.71               | 0                                                                                                                                                                                                         | 0.0899    | J1127, J1188                                                          | \$1,007,109                       | \$2,499,649                   |       |
|                                                                                                            |                     | 50                                                                                                                                                                                                        |                     | 50                                                                                                                                                                                                        |           |                                                                       |                                   |                               |       |

### **8.3 Network Upgrade Alternatives Considered**

All NRIS network upgrades identified in Table 8.1-2 are direct upgrades of the constraint facilities to ATC design standards and considered as least-cost solutions at this point. Therefore, no other alternatives were examined.

## **9.0 COST ALLOCATION**

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the System Impact Study report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

### **9.1 ERIS Network Upgrades Proposed for Apr 18 DPP Phase 1 Projects**

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the ERIS analysis. The ERIS network upgrades include thermal network upgrades and voltage support network upgrades identified in the steady-state analysis. For DPP April 2018 East (ATC) Phase 1 group, the total costs of ERIS network upgrades for the 2023 scenario are summarized in Tables 9.1-1, 9.1-2, 9.1-3, 9.1-4, 9.1-5, 9.1-6 and 9.1.7.

**Table 9.1-1 – ERIS Network Upgrades Identified in Steady State Analysis**

| Steady-State Injection Constraint                       | Facility Owner | Network Upgrade                                                                                 | ISD (For Cost Estimate Only) | Cost (\$) <sup>1,2,3</sup> |
|---------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------|------------------------------|----------------------------|
| Burlington – North Lake Geneva Tap 138 kV line          | ATC            | Burlington – North Lake Geneva Tap 138 kV line, reconductor <sup>4</sup>                        | 01/02/2021                   | 10,692,501                 |
| Sand Lake 138/69 kV transformer                         | ATC            | Sand Lake 138/69 kV transformer, replacement                                                    | 09/01/2020                   | 3,105,696                  |
| Wautoma 138/69 kV transformer                           | ATC            | Wautoma 138/59 kV transformer, replacement                                                      | 09/01/2020                   | 2,859,187                  |
| J986 POI – Port Edwards 138 kV line                     | ATC            | J986 POI – Port Edwards 138 kV line, uprate                                                     | 09/01/2020                   | 584,677                    |
| Concord – Rubicon 138 kV line                           | ATC            | Concord – Bark River 138 kV, new line; Concord – Crawfish River – Jefferson 138 kV line, uprate | 04/01/2021                   | 33,818,778                 |
| Summit 138 kV bus tie 2-3                               | ATC            |                                                                                                 |                              |                            |
| Cooney – Summit 138 kV line                             | ATC            |                                                                                                 |                              |                            |
| Concord – Cooney 138 kV line                            | ATC            |                                                                                                 |                              |                            |
| Saratoga 138/69 kV transformer                          | ATC            | Saratoga 138/115 kV transformer, re-tap CT                                                      | 09/01/2020                   | 114,149                    |
| Rocky Run – Werner West 345 kV line                     | ATC            | Rocky Run – Werner West 345 kV line, uprate                                                     | 09/01/2020                   | 330,135                    |
| North Appleton – Werner West 345 kV line                | ATC            | North Appleton – Werner West 345 kV line, uprate                                                | 09/01/2020                   | 597,741                    |
| Lublin – Lublin Pump 69 kV line                         | DPC            | Lublin – Lublin Pump 69 kV line, uprate                                                         | 09/01/2021                   | 364,865                    |
| Lublin Tap – Lublin Pump 69 kV line                     | DPC            | Lublin Tap – Lublin Pump 69 kV line, uprate                                                     | 09/01/2021                   | 4,135,135                  |
| Stone Lake 345/161 kV transformer                       | XCEL           | Stone Lake 345/161 kV transformer, replacement <sup>5</sup>                                     | 09/01/2021                   | 3,757,200                  |
| Overvoltage on J1085/J1121 POI – Stone Lake 345 kV line | ATC            | J1085/J1121 – Stone Lake 345 kV, install 80 Mvar line-connected reactor                         | 09/01/2021                   | 4,356,036                  |

<sup>1</sup> All Network Upgrades were estimated on the earliest ISD dollars of responsible generator.

<sup>2</sup> ATC Network Upgrades included a 10% contingency.

<sup>3</sup> No contingency was included for the Stone Lake transformer upgrade project according to Xcel Energy.

<sup>4</sup> Cost Estimate is the APR-18 ERIS cost (\$10,972,884) minus the AUG-17 NRIS cost (\$280,383) to uprate the line

<sup>5</sup> Cost Estimate is the APR-18 ERIS cost (\$7,655,700) minus the AUG-17 NRIS cost (\$3,898,500) to replace the transformer

**Table 9.1-2 – ERIS Network Upgrades in Dynamic Stability Analysis**

To be determined in MISO DPP 2018 April Phase 2

**Table 9.1-3 – ERIS Network Upgrades in Short Circuit Analysis**

To be determined in MISO DPP 2018 April Phase 2

**Table 9.1-4 – ERIS Affected System Network Upgrades**

To be determined in MISO DPP 2018 April Phase 2

**Table 9.1-5 – ERIS Network Upgrades Identified in PSCAD Analysis**

To be determined in MISO DPP 2018 April Phase 2

**Table 9.1-6 – ERIS PJM Affected System Network Upgrades**

To be determined in MISO DPP 2018 April Phase 2

**Table 9.1-7 – ERIS Shared Network Upgrades**

| Constraint | Facility Owner | Network Upgrade | Cost (\$) |
|------------|----------------|-----------------|-----------|
| None       | -              | -               | -         |

**9.2 NRIS Network Upgrades Proposed for Apr 18 DPP Phase 1 Projects**

Network upgrades for Network Resource Interconnection Service (NRIS) were identified in the MISO's deliverability analysis and listed in the Table 9.2-1 below.

**Table 9.2-1 – NRIS Network Upgrades Identified**

| Network Upgrade                                                                        | Facility Owner | ISD (For Cost Estimate Only) <sup>1</sup> | Cost Used for NRIS Cost Allocation (\$) <sup>2</sup> |
|----------------------------------------------------------------------------------------|----------------|-------------------------------------------|------------------------------------------------------|
| J1104 - Bark River 138 kV line, Reconductor                                            | ATC            | 09/01/2021                                | 3,479,462                                            |
| Germantown - Maple 138 kV line, Reconductor                                            | ATC            | 09/01/2021                                | 2,733,155                                            |
| Rock River - Marine 138 kV line, Reconductor                                           | ATC            | 01/02/2021                                | 2,499,649                                            |
| Darlington - Rock Branch 69 kV line, Uprate                                            | ATC            | 01/02/2021                                | 192,841                                              |
| North Monroe - Monticello Tap 69 kV line, Uprate                                       | ATC            | 01/02/2021                                | 41,915                                               |
| Bristol - Elkhorn 138 kV line, Uprate                                                  | ATC            | 01/02/2021                                | 193,170                                              |
| North Lake Geneva - North Lake Geneva Tap 138 kV line, Uprate                          | ATC            | 01/02/2021                                | 71,805                                               |
| Elkhorn - North Lake Geneva 138 kV line, Uprate <sup>3</sup>                           | ATC            | 01/02/2021                                | 114,514                                              |
| North Lake Geneva Tap - Burlington 138 kV line, Rebuild to Double Circuit <sup>4</sup> | ATC            | 01/02/2021                                | 6,804,658                                            |
| University - Mukwonago 138 kV line, Rebuild                                            | ATC            | 01/02/2021                                | 24,244,505                                           |
| Sugar Creek - North Lake Geneva Tap 138 kV line, Uprate                                | ATC            | 01/02/2021                                | 356,565                                              |
| Whitewater - Bluff Creek Tap - Sugar Creek 138 kV line, Uprate                         | ATC            | 01/02/2021                                | 350,275                                              |

<sup>1</sup> All Network Upgrades were estimated on the earliest ISD dollars of responsible generators.

<sup>2</sup> ATC Network Upgrades included a 10% contingency.

<sup>3</sup> Cost Estimate is the APR-18 NRIS cost (\$159,914) minus the AUG-17 NRIS cost (\$45,400) to uprate the line

<sup>4</sup> Cost Estimate is the APR-18 NRIS cost (\$17,777,542) minus the APR-18 ERIS cost (\$10,972,884) to reconductor the line minus the AUG-17 NRIS cost (\$280,383) to uprate the line

### 9.3 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst-case scenario for each specific generator project. Basically, whatever the highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left( \frac{\text{Max}(\text{Proj. A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj. i MW contribution on constraint})} \right)$$

### 9.4 Cost Estimating and Allocation Methodology for Short Circuit Upgrades

For each breaker shown to be loaded above 100% of rating a new breaker will be scoped and the cost of that upgrade will be assigned by to generators based on the MW impact provided from each generator for the worst case loading of the breaker.

For each substation that shows a bus having single-line-ground fault current increased by 10% or more, costs are assigned for ground grid upgrades at that substation. Only one cost estimate is scoped per substation regardless of the number of buses at that substation that show a 10% or greater increase in fault current. The largest fault current value at a substation will determine the ground grid upgrade costs. If the largest fault current is above 20 kA, the upgrades are assigned a planning level estimate of \$100,000 for equipment upgrades and \$50,000 is assigned for ground grid upgrades. If the largest fault current is below 20 kA, the upgrades are assigned a planning level estimate of \$50,000 for ground grid upgrades only.

Once costs are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all study generator contributing fault currents under the single line-ground short circuit fault simulation at the substation.

This methodology is applied to both ATC and non-ATC facilities modeled in ATC Protection models. For non-ATC facilities, it is interconnection customer's responsibility to work with facility owners to further refine costs, and

implementation of the mitigation projects (if needed). These are placeholder costs until facility studies are performed. Based on those results, costs are adjusted.

## **9.5 Cost Allocation Tables**

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the issued date of the System Impact Study report.

Assuming all generating facilities in the DPP 2018 April East (ATC) in group advance, Table 9.5-1, Table 9.5-2, Table 9.5-3, Table 9.5-4, Table 9.5-5, Table 9.5-6 and Table 9.5-7 show how the costs for ERIS steady-state Network Upgrades, short-circuit Network Upgrades, EMT stability Network Upgrades, PJM affected system Network Upgrades and NRIS Network Upgrades allocated to responsible generating facilities.

Table 9.5-1 – ERIS Steady-State Thermal Network Upgrade Costs Allocated to Each Generation Project, Part 1

| Required ERIS Network Upgrades                                                                  | Worst MW Impact and % Cost Allocation |                  |                |                  |                   |                   |                   |                   |                   |                |                  |               |                |                 |               | Total MW Impact |
|-------------------------------------------------------------------------------------------------|---------------------------------------|------------------|----------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|----------------|------------------|---------------|----------------|-----------------|---------------|-----------------|
|                                                                                                 | J986                                  | J1002            | J1003          | J1042            | J1044             | J1047             | J1051             | J1053             | J1085             | J1101          | J1121            | J1127         | J1129          | J1154           | J1188         |                 |
| Burlington – North Lake Geneva Tap 138 kV line, reconductor                                     | -                                     | -                | -              | 68.42 MW, 22.62% | 111.41 MW, 36.83% | 111.41 MW, 36.83% | -                 | -                 | -                 | -              | -                | 3.61 MW, 1.2% | 4.61 MW, 1.52% | -               | 3.00 MW, 1.0% | 302.46          |
| Sand Lake 138/69 kV transformer, replacement                                                    | 26.61 MW, 100%                        | -                | -              | -                | -                 | -                 | -                 | -                 | -                 | -              | -                | -             | -              | -               | -             | 26.61           |
| Wautoma 138/59 kV transformer, replacement                                                      | 29.00 MW, 51.83%                      | 26.95 MW, 48.17% | -              | -                | -                 | -                 | -                 | -                 | -                 | -              | -                | -             | -              | -               | -             | 55.95           |
| J986 POI – Port Edwards 138 kV line, uprate                                                     | 110.93 MW, 73.57%                     | 39.86 MW, 26.43% | -              | -                | -                 | -                 | -                 | -                 | -                 | -              | -                | -             | -              | -               | -             | 150.79          |
| Concord – Bark River 138 kV, new line; Concord – Crawfish River – Jefferson 138 kV line, uprate | -                                     | -                | 6.51 MW, 1.01% | -                | -                 | -                 | 102.93 MW, 15.95% | 500.18 MW, 77.49% | -                 | -              | -                | -             | -              | 35.83 MW, 5.55% | -             | 645.45          |
| Saratoga 138/115 kV transformer, re-tap CT                                                      | 19.60 MW, 100%                        | -                | -              | -                | -                 | -                 | -                 | -                 | -                 | -              | -                | -             | -              | -               | -             | 19.60           |
| Rocky Run – Werner West 345 kV line, uprate                                                     | 16.16 MW, 10.6%                       | -                | -              | -                | -                 | -                 | -                 | -                 | 98.63 MW, 65%     | 4.22 MW, 2.8%  | 32.88 MW, 21.6%  | -             | -              | -               | -             | 151.88          |
| North Appleton – Werner West 345 kV line, uprate                                                | 16.74 MW, 10.16%                      | -                | -              | -                | -                 | -                 | -                 | -                 | 107.76 MW, 65.39% | 4.37 MW, 2.65% | 35.92 MW, 21.8%  |               | -              | -               | -             | 164.79          |
| Lublin – Lublin Pump 69 kV line, uprate                                                         | -                                     | -                | -              | -                | -                 | -                 | -                 | -                 | 2.53 MW, 75%      | -              | 0.84 MW, 25%     | -             | -              | -               | -             | 3.37            |
| Lublin Tap – Lublin Pump 69 kV line, uprate                                                     | -                                     | -                | -              | -                | -                 | -                 | -                 | -                 | 2.53 MW, 75%      | -              | 0.84 MW, 25%     | -             | -              | -               | -             | 3.37            |
| Stone Lake 345/161 kV transformer, replacement                                                  | -                                     | -                | -              | -                | -                 | -                 | -                 | -                 | 134.54 MW, 59.83% | -              | 90.34 MW, 40.17% | -             | -              | -               | -             | 224.89          |
| J1085/J1121 – Stone Lake 345 kV, install 80 Mvar line-connected reactor                         | -                                     | -                | -              | -                | -                 | -                 | -                 | -                 | 300 MW, 60%       | -              | 200 MW, 40%      | -             | -              | -               | -             | 500             |

Table 9.5-1 – ERIS Steady-State Thermal Network Upgrade Costs Allocated to Each Generation Project, Part 2

| Required ERIS Network Upgrades                                                                  | Cost Allocation (\$) |           |         |           |           |           |           |            |             |        |             |         |         |           |         |
|-------------------------------------------------------------------------------------------------|----------------------|-----------|---------|-----------|-----------|-----------|-----------|------------|-------------|--------|-------------|---------|---------|-----------|---------|
|                                                                                                 | J986                 | J1002     | J1003   | J1042     | J1044     | J1047     | J1051     | J1053      | J1085       | J1101  | J1121       | J1127   | J1129   | J1154     | J1188   |
| Burlington – North Lake Geneva Tap 138 kV line, reconductor                                     | -                    | -         | -       | 2,418,767 | 3,938,539 | 3,938,539 | -         | -          | -           | -      | -           | 127,717 | 162,919 | -         | 106,020 |
| Sand Lake 138/69 kV transformer, replacement                                                    | 3,105,696            | -         | -       | -         | -         | -         | -         | -          | -           | -      | -           | -       | -       | -         | -       |
| Wautoma 138/59 kV transformer, replacement                                                      | 1,481,952            | 1,377,235 | -       | -         | -         | -         | -         | -          | -           | -      | -           | -       | -       | -         | -       |
| J986 POI – Port Edwards 138 kV line, uprate                                                     | 430,135              | 154,542   | -       | -         | -         | -         | -         | -          | -           | -      | -           | -       | -       | -         | -       |
| Concord – Bark River 138 kV, new line; Concord – Crawfish River – Jefferson 138 kV line, uprate | -                    | -         | 341,097 | -         | -         | -         | 5,392,870 | 26,207,430 | -           | -      | -           | -       | -       | 1,877,380 | -       |
| Saratoga 138/115 kV transformer, re-tap CT                                                      | 114,149              | -         | -       | -         | -         | -         | -         | -          | -           | -      | -           | -       | -       | -         | -       |
| Rocky Run – Werner West 345 kV line, uprate                                                     | 35,131               | -         | -       | -         | -         | -         | -         | -          | 214,376     | 9,169  | 71,459      | -       | -       | -         | -       |
| North Appleton – Werner West 345 kV line, uprate                                                | 60,721               | -         | -       | -         | -         | -         | -         | -          | 390,874     | 15,854 | 130,291     | -       | -       | -         | -       |
| Lublin – Lublin Pump 69 kV line, uprate                                                         | -                    | -         | -       | -         | -         | -         | -         | -          | 273,649     | -      | 91,216      | -       | -       | -         | -       |
| Lublin Tap – Lublin Pump 69 kV line, uprate                                                     | -                    | -         | -       | -         | -         | -         | -         | -          | 3,101,351   | -      | 1,033,784   | -       | -       | -         | -       |
| Stone Lake 345/161 kV transformer, replacement                                                  | -                    | -         | -       | -         | -         | -         | -         | -          | 2,247,824   | -      | 1,509,376   | -       | -       | -         | -       |
| J1085/J1121 – Stone Lake 345 kV, install 80 Mvar line-connected reactor                         | -                    | -         | -       | -         | -         | -         | -         | -          | 2,613,622   | -      | 1,742,414   | -       | -       | -         | -       |
| Total ERIS Network Upgrade Cost (\$) Allocated To Each Generator                                | 5,227,784            | 1,531,777 | 341,097 | 2,418,767 | 3,938,539 | 3,938,539 | 5,392,870 | 26,207,430 | \$8,841,697 | 25,023 | \$4,578,540 | 127,717 | 162,919 | 1,877,380 | 106,020 |

**Table 9.5-2 – ERIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project**  
To be determined in MISO DPP 2018 April Phase 2

**Table 9.5-3 – ERIS EMT Stability Network Upgrade Costs Allocated to Each Generation Project**  
To be determined in MISO DPP 2018 April Phase 2

**Table 9.5-4 – PJM Affected System Costs Allocated to Each Generation Project**  
To be determined in MISO DPP 2018 April Phase 2

Table 9.5-5 – NRIS Network Upgrade Costs Allocated to Each Generation Project, Part 1

| Required NRIS Network Upgrades                                                                                                                                                                         | Worst MW Impact and % Cost Allocation Details Provided by MISO |                     |                      |                      |                      |                   |                    |                     |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------|----------------------|----------------------|----------------------|-------------------|--------------------|---------------------|--------------------|
|                                                                                                                                                                                                        | J1010                                                          | J1011               | J1042                | J1044                | J1047                | J1104             | J1127              | J1129               | J1188              |
| J1104 - Bark River 138 kV line, Reconductor                                                                                                                                                            | -                                                              | -                   | -                    | -                    | -                    | 100%              | -                  | -                   | -                  |
| Germantown - Maple 138 kV line, Reconductor                                                                                                                                                            | -                                                              | -                   | -                    | -                    | -                    | 100%              | -                  | -                   | -                  |
| Rock River - Marine 138 kV line, Reconductor                                                                                                                                                           | -                                                              | -                   | -                    | -                    | -                    | -                 | 6.67 MW,<br>59.71% | -                   | 4.5 MW,<br>40.29%  |
| Darlington - Rock Branch 69 kV line, Uprate                                                                                                                                                            | -                                                              | -                   | -                    | -                    | -                    | -                 | 100%               | -                   | -                  |
| North Monroe - Monticello Tap 69 kV line, Uprate                                                                                                                                                       | -                                                              | -                   | -                    | -                    | -                    | -                 | 100%               | -                   | -                  |
| Bristol - Elkhorn 138 kV line, Uprate                                                                                                                                                                  | -                                                              | -                   | -                    | -                    | -                    | -                 | 3.56 MW,<br>33.94% | 4.38 MW ,<br>41.75% | 2.55 MW,<br>24.31% |
| North Lake Geneva - North Lake Geneva Tap 138 kV line, Uprate                                                                                                                                          | -                                                              |                     | 112.6 MW,<br>29.67%  | 127.12 MW,<br>33.49% | 127.12 MW,<br>33.49% | -                 | 4.19 MW,<br>1.10%  | 5.59 MW,<br>1.47%   | 2.95 MW,<br>0.78%  |
| Elkhorn - North Lake Geneva 138 kV line, Uprate                                                                                                                                                        | -                                                              | -                   | -                    | -                    | -                    | -                 | 5.92 MW,<br>32.62% | 7.6 MW ,<br>41.87%  | 4.63 MW,<br>25.51% |
| University - Mukwonago 138 kV line, Rebuild                                                                                                                                                            | -                                                              | -                   | 28.19 MW,<br>25.44%  | 35.49 MW,<br>32.01%  | 35.49 MW,<br>32.01%  | -                 | 3.75 MW<br>3.38%   | 4.7 MW,<br>4.24%    | 3.24 MW,<br>2.92%  |
| North Lake Geneva Tap - Burlington 138 kV line, Rebuild to Double Circuit<br>Sugar Creek - North Lake Geneva Tap 138 kV line, Uprate<br>Whitewater - Bluff Creek Tap - Sugar Creek 138 kV line, Uprate | 70.16 MW,<br>11.01%                                            | 70.16 MW,<br>11.01% | 130.34 MW,<br>20.47% | 167.51 MW,<br>26.29% | 167.51 MW,<br>26.29% | 5.02 MW,<br>0.79% | 8.5 MW,<br>1.33%   | 10.85 MW,<br>1.7%   | 7.04 MW,<br>1.11%  |

Table 9.5-5 – NRIS Network Upgrade Costs Allocated to Each Generation Project, Part 2

| Required NRIS Network Upgrades                                                                                                                                                                         | Cost Allocation (\$) |         |           |           |           |           |           |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|                                                                                                                                                                                                        | J1010                | J1011   | J1042     | J1044     | J1047     | J1104     | J1127     | J1129     | J1188     |
| J1104 - Bark River 138 kV line, Reconductor                                                                                                                                                            | -                    | -       | -         | -         | -         | 3,479,462 | -         | -         | -         |
| Germantown - Maple 138 kV line, Reconductor                                                                                                                                                            | -                    | -       | -         | -         | -         | 2,733,155 | -         | -         | -         |
| Rock River - Marine 138 kV line, Reconductor                                                                                                                                                           | -                    | -       | -         | -         | -         | -         | 1,492,541 | -         | 1,007,109 |
| Darlington - Rock Branch 69 kV line, Uprate                                                                                                                                                            | -                    | -       | -         | -         | -         | -         | 192,841   | -         | -         |
| North Monroe - Monticello Tap 69 kV line, Uprate                                                                                                                                                       | -                    | -       | -         | -         | -         | -         | 41,915    | -         | -         |
| Bristol - Elkhorn 138 kV line, Uprate                                                                                                                                                                  | -                    | -       | -         | -         | -         | -         | 65,562    | 80,648    | 46,960    |
| North Lake Geneva - North Lake Geneva Tap 138 kV line, Uprate                                                                                                                                          | -                    | -       | 21,305    | 24,047    | 24,047    | -         | 790       | 1,056     | 560       |
| Elkhorn - North Lake Geneva 138 kV line, Uprate                                                                                                                                                        | -                    | -       | -         | -         | -         | -         | 37,355    | 47,947    | 29,213    |
| University - Mukwonago 138 kV line, Rebuild                                                                                                                                                            | -                    | -       | 6,167,802 | 7,760,666 | 7,760,666 | -         | 819,464   | 1,027,967 | 707,940   |
| North Lake Geneva Tap - Burlington 138 kV line, Rebuild to Double Circuit<br>Sugar Creek - North Lake Geneva Tap 138 kV line, Uprate<br>Whitewater - Bluff Creek Tap - Sugar Creek 138 kV line, Uprate | 827,016              | 827,016 | 1,537,604 | 1,974,773 | 1,974,773 | 59,341    | 99,903    | 127,695   | 83,378    |
| Total NRIS Network Upgrade Cost(\$)<br>Allocated To Each Generator                                                                                                                                     | 827,016              | 827,016 | 7,726,710 | 9,759,486 | 9,759,486 | 6,271,958 | 2,750,371 | 1,285,314 | 1,875,158 |

## **10.0 AVAILABLE APPENDIX DOCUMENTS (NOT ATTACHED)**

Appendix A – Study Criteria, Methodology, and Assumptions

Appendix B – ATC Planning Criteria and Generation Facility Interconnection Guide

Appendix C – Interconnection Facility Project Diagrams and Modeling Details

Appendix D – Network Upgrade Project Diagrams

Note: Project Diagrams were not developed for line uprate projects.

Appendix E – Steady State Power Flow Results

Appendix I – MISO Deliverability Study Results

Appendix J – Assessed System Performance Reference

Appendix K – J1085/J1121 Additional Studies